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Deformation and transformation of rock-forming minerals by natural and experimental shock processes

I. Behavior of minerals under shock compression *

By

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With 36 Figures and 6 Tables

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Zusammenfassung: Es wird ein Überblick über die physikalischen Grundlagen der dynamischen Kompression von Festkörpern durch Stoßwellen unter besonderer Berücksichtigung gesteinsbildender Mineralien gegeben. Die wesentlichen experimentellen Methoden zur Bestimmung der Zustandsdaten unter hohen dynamischen Drucken und zur Rückgewinnung der durch Stoßwellen beanspruchten Mineralien für nachfolgende physikalische und mineralogische Untersuchungen von irreversiblen Stoßwelleneffekten werden besprochen. Die besonderen Bedingungen der Stoßwellenmetamorphose in Meteoritenkratern werden kurz diskutiert. Danach folgt eine systematische Erörterung der wichtigsten durch Stoßwellendrucke bis etwa 1 megabar in Mineralien hervorgerufenen Deformations- und Transformationsprozesse. Die aus Labor-Experimenten und Nuklearexplosionen, sowie aus der Untersuchung von terrestrischen, lunaren und meteoritischen Impakt-Formationen gewonnenen Daten werden diskutiert und referiert.

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Abstract: An introduction to the basic conditions of the shock wave compression of solids with special regard to rock-forming minerals is given. Experimental techniques used for the determination of equation-of-state data as well as for recovery experiments for subsequent studies of irreversible shock effects in minerals are discussed. The special conditions of shock metamorphism in impact craters are briefly described. The main part of this review is devoted to a description of the basic deformation and transformation processes in shocked minerals at pressures up to about 1 megabar. Data from laboratory shock experiments, from nuclear explosions, and from terrestrial, lunar, and meteoritic impact formations are discussed and referenced.

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1. Introduction

Some 15 years ago geologists generally were convinced that any kind of natural metamorphism of rocks and minerals must have its origin in endogenic processes in which mineralogical and structural adjustments of solid rocks take place in response to changes in physical and chemical conditions imposed at some depth below the surface. Increasing space exploration and consequent study of meteorites and terrestrial impact structures coupled with the increasing experimental capabilities of shock wave studies since 1955 have drawn and continue to draw the attention of geologists to shock-induced metamorphic processes associated with terrestrial impact structures as well as meteorites. The accomplishments of those studies could only be achieved by mutual stimulation of very different research fields such as shock physics, dynamic and static ultrahigh pressure epxerimentation, geology and petrology of terrestrial impact craters, photogeology of the lunar surface explored by space vehicles, experimental hypervelocity impact cratering, research on meteorites, and astrophysics. Consequently, our knowledge about the reaction of geological materials against the action of strong shock waves has very rapidly increased in the past decade. However, it has also forced geologists to familiarize themselves with unconventional and new physical processes and experimental techniques in order to understand their observations in nature. This change in the range of conservative geological ideas and activities is part of a more general metamorphosis of "geology" into "planetology" which is more than just a new term in geological science but is the expression of a more comprehensive research field characterized by the common efforts of a wide variety of disciplines.

The theory of shock waves dates back to the last century when STOKES (1848) and RANKINE (1870) established the basic mathematical treatment of shock waves. The first experimental application began during the Second World War (SCHALL, 1944), when shock waves became important in military and industrial technology. Since that time shock wave theory and experimentation has become a powerful and sophisticated tool in materials science

which has yielded fundamental data about the thermodynamic state of solids at ultrahigh pressures and temperatures. Pressures as high as almost 10 megabar (KORMER et al., 1962) have been experimentally produced by laboratory high explosive devices and even higher pressures are achieved by underground nuclear explosions or, in nature, by the impact of giant meteoroids. Shock wave techniques have also provided considerable new data in the geophysics of the internal structure of planets mainly in terms of data on possible high pressure phase transitions. However, the gain in the extension of the p-T-field which now can be experimentally covered, has to be paid by the extreme short duration of the peak pressure, which is in the order of microseconds, and by the associated temperature increase which becomes severe at very high pressures, e.g., some 10,000 degrees at some megabars in the case of solid materials.

A number of excellent review articles on the physics and experimental techniques of shock waves have been written by several experts in this field (COURANT & FRIEDRICHS, 1948; RICE et al., 1958; DEAL, 1962; DUVALL, 1962; DUVALL & FOWLES, 1963; MCQUEEN, 1963; ALDER, 1963; DORAN & LINDE, 1966; HAMANN, 1966; ZEL'DOVICH & RAIZER, 1967; DUVALL, 1968; MCQUEEN et al., 1970; ZHARKOV & KALININ, 1971), and the reader is referred to those articles for any detailed information.

Shock fronts may produce transient as well as residual changes when passing a solid. Transient and permanent shock effects are of considerable interest in materials science and have been experimentally studied in a wide variety of solids (see DORAN & LINDE, 1966). The main interest of geologists is devoted to permanent shock effects in rocks and minerals and their formation conditions. Since 1960 a great number of deformation and transformation phenomena in rock-forming minerals produced by natural and artificial shock processes have been reported. A first summary of data may be obtained from a collection of articles published in the Proceedings Volumes of two conferences especially devoted to the study of shock metamorphism and impact cratering (FRENCH & SHORT, 1968; HÖRZ, 1971). In spite of a considerable compilation of data in the literature, we are still at the beginning of an actual physical understanding of shock-induced deformation and transformation mechanisms and much more shock experimentation at well-defined pressure-temperature-time conditions is necessary, which should be combined with more intense studies of the relations between the shock wave structure and the material's post shock properties and structural state.

Unfortunately, no systematic review on the behavior of rock-forming minerals under shock compression summarizing theoretical and experimental shock data as well as observations from natural shock processes is available so far. Much valuable data is widely scattered through the literature or even unpublished, or buried in hardly discoverable technical notes or reports. Some confusion about terminology as well as interpretation has arisen already which is especially evident from the interpretation of shocked lunar rocks and minerals from the study of which a flood of observations has been obtained in the last two years (see LEVINSON, 1970, 1971; KING, 1972). It is the purpose of this article to assemble those widely scattered data as far as possible and to attempt to describe and interpret the effects of shock waves

on minerals. This review does not claim to be comprehensive and omissions will certainly occur. Especially, some experimental shock techniques and the particular mode of shock deformation of minerals in experimental impact and explosive cratering events have been discussed only briefly and probably insufficiently. This is also true for the implications of shock effects in minerals in terms of progressive shock metamorphism of terrestrial, lunar, and meteoritic rock materials. Also studies of the plastic deformation of shocked minerals by modern X-ray and electron microscopic methods have not been discussed in detail because extensive research in those fields is just at the beginning.

The present article will be divided into two parts. Part I describes the basic principles of shock waves in solids, the experimental techniques and the deformation and transformation behavior of minerals under shock compression. Part II (STÖFFLER, 1972) will deal with the physical properties of shocked minerals in the various stages of shock damage.

2. Some principles of shock wave propagation in solids

2.1. Basic Physics of Shock Waves

A stress wave propagating with supersonic velocity is formed when the free surface of a solid body is suddenly accelerated, e.g., by the impact of a projectile or by a chemical or nuclear explosion. Since the compressibility of a solid generally decreases with increasing pressure the stress wave will immediately steepen to a shock wave — also called shock front — which is a discontinuity in pressure, density, and internal energy. The constituent particles are accelerated and the material moves behind the shock front with a certain particle velocity. In practice, a shock front has a finite extent and is followed by a rarefaction wave which travels faster than the shock front. This results from the fact that there is a positive particle velocity behind the shock front that increases the sound velocity in the compressed material. The rarefaction wave gradually overtakes the shock wave causing a decrease of the peak pressure and broadening of the entire shock transition with increasing distance of propagation.

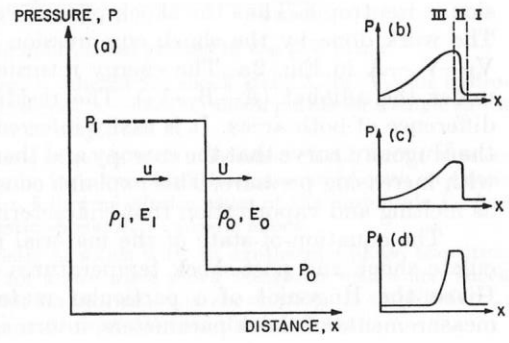


Fig. 1. Pressure-distance profiles for shock fronts (see text)

a) uniform shock at hydrodynamic conditions
b)—d) non-uniform shocks:
b) see text
c) shock front induced by detonation of high-explosive in contact
d) shock front induced by the impact of a flying plate.

For the mathematical treatment of a shock transition the shock wave is considered to be steady and of infinite plane geometry connecting a uniform, undisturbed state with an uniform shocked state. Assuming that the material behaves hydrodynamically under compression the basic equations of the conservation of momentum, mass and energy can be applied describing the thermodynamic condition across the shock transition sometimes referred to as "jump conditions."

Consider a plane shock wave traveling with a velocity U into a material initially at rest and then accelerated to a particle velocity u. The density ϱ_o, the pressure P_o and internal energy E_o rise to ϱ_1, P_1, and E_1 (Fig. 1a). From the conservation of mass, we get

$$\varrho_o U = \varrho_1 (U - u). \tag{1}$$

The conservation of momentum gives

$$P_1 - P_o = \varrho_o U u \tag{2}$$

and the conservation of energy is expressed by

$$P_1 u = \frac{1}{2} \varrho_o U u^2 + \varrho_o U (E_1 - E_o). \tag{3}$$

Eliminating U and u by combining equations (1) to (3), gives

$$E_1 - E_o = \frac{1}{2}(P_1 + P_o)(V_o - V_1) \tag{4}$$

where V is the specific volume $\frac{1}{\varrho}$. This equation is called the Rankine-Hugoniot relation. In other words, this relation describes the locus of all shock states (P_1, V_1, E_1) achievable by shock waves of various intensity in a particular solid of the initial state P_o, V_o, E_o. If graphically represented in the pressure-volume plane this locus is commonly referred to as Hugoniot curve (Fig. 2). A detailed derivation of the above equations is given by Duvall & Fowles (1963) for example.

Inspection of the Hugoniot curve shows that irreversible work is done in a shock process resulting in a certain amount of post-shock heat. The thermodynamic path for a material shocked to the final state P_1, V_1, E_1 (Fig. 2a) is represented by the Raleigh-line (line V_oA). The rarefaction brings the compressed material back to ambient conditions adiabatically. The compression phase is a nonisentropic process, whereas the adiabatic expansion is isentropic. Thus the shock process results in an increase in entropy. The work done by the shock compression equals the area of the triangle V_1—V_o—A in Fig. 2a. The energy returned on the expansion is the area under the adiabat (A—B—V_1). The residual heat is proportional to the difference of both areas. It is easily inferred from the upward concavity of the Hugoniot curve that the entropy and therefore the residual heat increases with increasing pressure. This explains conspicuous post shock effects such as melting and vaporization that characterize hypervelocity impact events.

The equation-of-state of the material must be known in order to calculate shock and post-shock temperatures of shock compressed materials. Given the Hugoniot of a particular material obtained by experimental measurements of shock parameters, information about the equation-of-state

can be obtained via the Mie-Grüneisen-equation. If the Grüneisen ratio ($\gamma = V(\partial P/\partial E)$) and the specific heat at constant volume is known, temperatures and other thermodynamic parameters can be calculated for any P, V state. The basic relations for the computation of shock temperatures along a Hugoniot curve are discussed in detail by several review articles, e.g., by McQueen (1963). Calculation of post-shock temperatures is based on the knowledge of the expansion isentropes. Since precise data on equation of

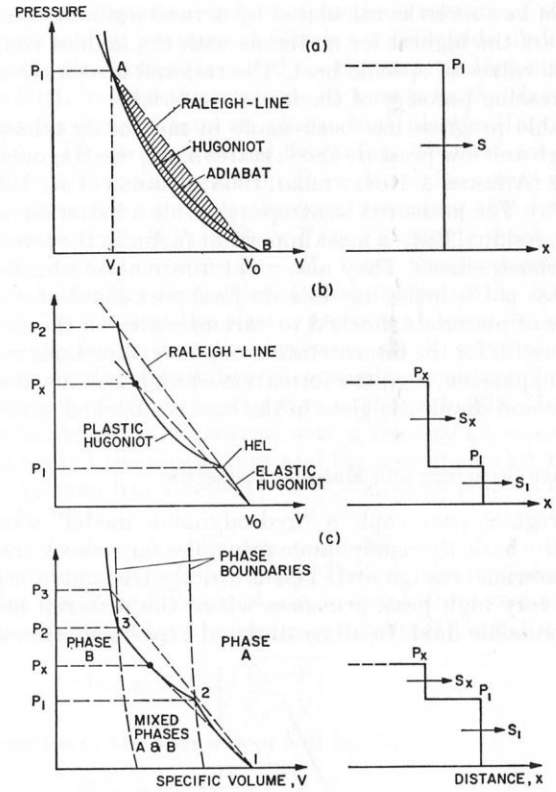


Fig. 2. Generalized Hugoniot curves plotted in the p-V plane

a) Left: Hugoniot curve and expansion adiabat for shock state A (p_1, V_1) for hydrodynamic conditions ("hydrodynamic model"). — Right: shock profile for a single shock with an amplitude P_1.

b) Left: Hugoniot curve characteristic of solids with a dynamic elastic limit ("elastic-plastic model"). — Right: Shock profile for a two wave shock transition with an elastic percursor of the amplitude P_1 and a following plastic wave of the amplitude P_x (see text); S_1 = elastic shock, S_x = plastic shock.

c) Left: Hugoniot curve characteristic of solids with an exothermic phase transition (see text). — Right: Shock profile for a two wave shock transition with a first shock wave of amplitude P_1 and a second, transformational shock wave of amplitude P_x; S_1 = plastic shock, S_x = transformational shock (see text).

state and hence expansion isentropes are not available for many geological materials, an estimate for the residual heat can be made by approximating the expansion isentrope with the Hugoniot (Gault & Heitowit, 1963) or more exactly by determining experimentally the shape of the expansion isentrope (release adiabat, see Ahrens & Rosenberg, 1968; Ahrens et al., 1969a) and therefrom calculating the gain in internal energy. In the former case the hatched area of Fig. 2a is taken as an approximate representation of the residual heat ("waste heat"). Fig. 3 shows a review of data on shock and post-shock temperatures of some of the most common rock-forming minerals and of two rocks as calculated by several authors. Generally, these temperatures are the highest for materials with the highest compressiblities and the lowest values of specific heat. The residual heat increases considerably with increasing porosity of the initial material.

Considerable progress has been made in measuring release isentropes centered at high and low pressure shock states along the Hugoniot using various techniques (Ahrens & Rosenberg, 1968; Ahrens et al., 1969a; Petersen et al., 1970). The measured isentropes provide a better basis for the calculation of the residual heat, a most important factor in the formation of permanent post-shock effects. They also yield information about the thermodynamic release paths including data on final post-shock states, e.g., post-shock volumes of materials shocked to various states on the Hugoniot. The latter is most useful for the interpretation of phase transitions occurring during adiabatic expansion, e.g., the formation of high and low density phases such as coesite and diaplectic glass in the case of shocked quartz.

2.2. Shock Wave Structure and Material Properties

In the foregoing paragraph, a "hydrodynamic model" was used in the derivation of the basic thermodynamic principles for a shock transition. The strength of materials was ignored. This is strictly true only for steady-state conditions at very high peak pressures where the material behaves as an inviscid, compressible fluid. In all natural and experimental shock processes

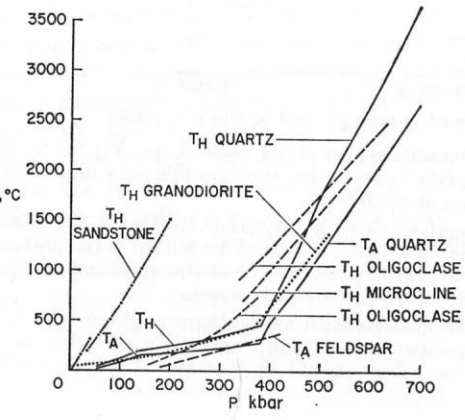


Fig. 3. Shock pressure versus shock temperature T_H and post-shock temperature T_A for framework silicates, granodiorite (Borg, 1972), and sandstone (Ahrens & Gregson, 1964). Data for feldspars are taken from Ahrens et al. (1969a), data for quartz from Wackerle (1962).

however we deal with "unsteady" or "non-uniform" conditions as shown in Fig. 1b—d, which illustrates typical shock profiles. The profiles are divided into three regions (see Duvall & Fowles, 1963). Only region I is steady (Fig. 1b). Regions II and III are unsteady due to the rarefaction wave which changes regions II (shock transition) and III with time and distance of propagation. Moreover, from the experimentation at low shock pressures and from the observed post-shock deformations in materials shocked to pressure in the neighborhood of the dynamic yield strength it is quite obvious that material properties such as the yield point or high pressure phase transitions have an important influence on the structure of the shock front.

The critical shock pressure at which a solid yields under the uniaxial strain of a plane shock wave is called the dynamic or Hugoniot elastic limit (HEL). The HEL implies the presence of a cusp in the stress-strain curve (see Fig. 2b) dividing it into an elastic and plastic Hugoniot: "elastic-plastic model". It can be inferred from Fig. 2b that for all final peak pressures P_x fulfilling the condition $P_1 < P_x < P_2$ a two-wave structure exists which gives rise to an elastic wave preceding the shock wave, now called the plastic wave. If $P_x < P_1$ or $P_x > P_2$ a single shock front is stable. It is called the elastic wave for $P_x < P_1$. Since the propagation velocity of a shock wave is directly proportional to the slope of the Raleigh-line it is obvious from Fig. 2b that the elastic precursor having an amplitude P_1 is always traveling faster than the following shock wave of amplitude P_x. The condition for the instability of a single shock can be expressed as follows: Assuming a two wave structure as shown in Fig. 2b the first shock with an amplitude of $P_1—P_o$ travels with a velocity $U_1—u_1$ with respect to the material behind the shock front and the second shock having an amplitude of $P_x—P_1$ then has a velocity of $U_x—u_1$ with respect to the material ahead of it. If

$$U_x—u_1 < U_1—u_1$$

the two wave structure will be stable. That is, the first shock will not be overtaken by the second shock. A single shock front can exist. The term U—u can be calculated by combining equations (1) and (2):

$$U—u = V_1\sqrt{\frac{P_1—P_o}{V_o—V_1}} \tag{5}$$

and the velocities of the two waves will be

$$U_x—u_1 = \sqrt{\frac{P_x—P_1}{V_1—V_x}}$$

$$\text{and } U_1—u_1 = \sqrt{\frac{P_1—P_o}{V_o—V_x}}$$

and finally the condition for the instability of a single shock front will be (Rice et al., 1958)

$$\frac{P_x—P_1}{V_1—V_x} < \frac{P_1—P_o}{V_o—V_1}.$$

A two-wave structure may also be caused by a phase transition. This is true for the hydrodynamic as well as the elastic-plastic model. This case

is illustrated in Fig. 2c. The cusp of the Hugoniot at P_1 is caused by the phase transition and the same conditions for the development of a two wave structure hold true as shown above for the HEL. Most characteristic for many materials of geologic interest, e.g., silicates, is a region of high compressibility representing the so-called mixed-phase or two-phase regime of the Hugoniot which extends to a second discontinuity at 3 (Fig. 2c) at which the transformation to a high-pressure-phase is considered to be complete. At peak pressures between 2 and 3 the transformation is only partially complete. The two-phase region will be the broader the more sluggish the phase transformation takes place. All final shock states above 3 represent states of the high-pressure-phase along the so-called high-pressure-phase Hugoniot. Since most solids exhibit a dynamic elastic limit in addition to one or more phase transitions, the shock wave may split into a multiple wave structure: elastic percursor, plastic wave and one or more transformational waves.

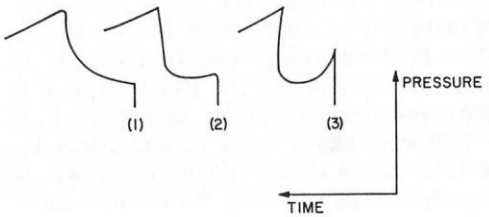


Fig. 4. Measured stress-time profiles for a two wave front in
a) aluminum
b) quartzite
c) germanium and silicon showing stress relaxation behind the elastic percursor (after Doran & Linde, 1966).

The interpretation of a multiple shock front in terms of material properties (elastic limit, phase transitions) is very often complicated by time and rate effects such as stress relaxation or viscosity. Hence multiple wave fronts are usually not as ideal as those shown in the profiles of Fig. 2a and b. They may be appreciably modified as shown by some examples of experimentally determined stress-time profiles (Fig. 4). Some materials show a pronounced decrease of stress behind the elastic wave front. Moreover the amplitude of the elastic wave decreases with propagation distance which is certainly due to stress relaxation effects (see Duvall & Fowles, 1963). The wave structure may be further complicated by strain rate effects which tend to thicken the shock transition zone. These effects probably have a significant influence on the kind of plastic deformation leading to particular post-shock deformation patterns. The relations between certain shock wave structures and particular post-shock effects are not yet well understood, however, especially in the case of silicates. Probably a complete understanding of these relationships will not be achieved until models describing the shock wave propagation in terms of crystal lattices (Tsai & Beckett, 1966; Manvi & Duvall, 1969) and of the motion of crystal lattice defects are available. Dislocation models of the shock front have been developed only for simple crystal structures of metallurgical interest (see e.g. discussion in Doran & Linde, 1966, Hornbogen, 1962, Johnson et al., 1970).

3. Experimental techniques

The aim of all experimental shock wave techniques is to measure the main thermodynamic quantities of a shock transition such as pressure and volume. Because these quantities cannot be determined directly, they must be calculated via the Hugoniot equations (equations (1) and (2)) from velocity measurements, e.g., from particle velocity u and shock velocity U. This procedure leads to the determination of the Hugoniot curve and finally to the equation-of-state (e.g., entropy, shock temperature). However, the experimental technique destroys the specimen and does not provide the possibility of a study of irreversible effects in the shocked specimen. Special recovery techniques must be applied in order to study permanent shock effects. Since the main purpose of this article is to present some results of post-shock studies, the recovery techniques will be discussed somewhat more in detail.

3.1. Generation of Plane Shock Waves

The generation of a plane shock wave is desirable because calculation of thermodynamic quantities is based on the model of a plane wave shock transition as described above. Plane waves can either be achieved by explosive systems in contact with the specimen or by hypervelocity plane impact of solid plates on the specimen using explosive systems as well as gun techniques.

Some explosive systems, commonly called "plane-wave-generators" are shown in Fig. 5 (Duvall & Fowles, 1963). The most commonly used two-explosive lens consists of a cone of explosive of high detonation velocity surrounding an inner cone of an explosive with low detonation velocity. The appropriate cone angle is determined by the ratio of the detonation velocities (Deal, 1962). Most explosives used for this purpose are mixtures of TNT with RDX, HMX or PETN (for details see Cook, 1958; Deal, 1962).

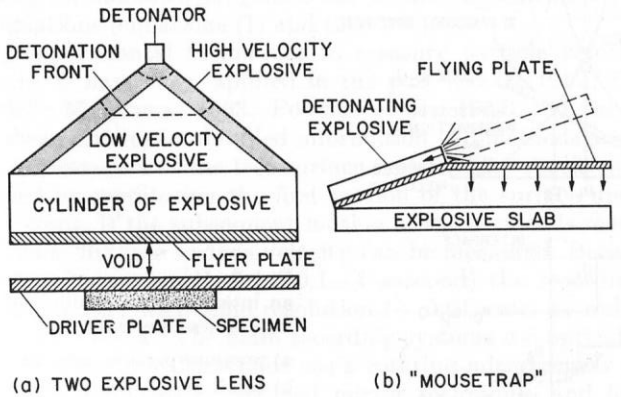


Fig. 5. Most commonly used high explosive plane-wave-generators (see text).

Plane wave generators are used either in direct contact with a driver plate and the specimen or by accelerating a flyer plate impacting the plane specimen. The latter method, called "free run technique", provides higher peak pressures than the contact method. By changing such parameters as the density and thickness of the metal plate and the kind of explosive the final peak pressure achieved in the specimen can be varied and is commonly in the 100—1,500 kbar range in the case of driver plates and in the 300—10,000 kbar range for flying plates depending on the density of the specimen. Gun devices (light-gas or explosively driven), that generate plane shock waves have been improved appreciably in recent years (JONES et al., 1966; CABLE, 1970; FOWLES et al., 1970). They are especially useful in the low pressure range (< 300 kbar), particularly below 100 kbar.

3.2. Hugoniot Measurements

Let us first consider the physical conditions of the transmission of a shock wave at a plane interface, e.g., metal plate-specimen interface, referring to the plane wave geometry described above. A representation of the known Hugoniot curves of the driver plate and the specimen in the pressure-particle-velocity plane as shown in Fig. 6a and b may be used to explain the interface conditions. If a shock wave of peak pressure P_1 and particle velocity u_1 propagating from left to right in the driver metal plate, represented by the Hugoniot curve I in Fig. 6a hits an interface to another material, the specimen, two wave fronts will arise from the interface. One is transmitted forward into the specimen and another one is reflected back into the metal plate. The locus of the shock states achieved by the reflected wave in the metal plate is given by the mirror curve of Hugoniot I intersecting Hugoniot I at the state P_1, u_1, and reflected at a vertical line passing through state P_1, u_1. This wave may either be a rarefaction or a shock wave depending on the density or, more exactly, on the shock impedance of the

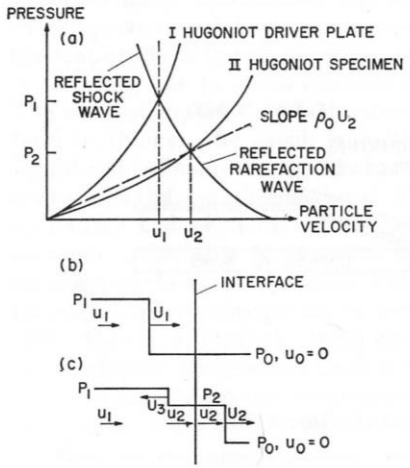


Fig. 6. Interaction of a shock wave at an interface to a medium of lower shock impedance (see text)
a) pressure-particle velocity plot
b) shock profile before interaction
c) shock profile after interaction

specimen (initial density times shock velocity, $\rho_0 U$). Assuming a lower shock impedance for the specimen, given by the slope of the dashed straight line in Fig. 6a, the reflected wave is a rarefaction wave. This rarefaction brings the metal plate from a state P_1, u_1, to a lower peak pressure P_2 thereby increasing the particle velocity to u_2. Since pressure and particle velocity must be continuous across the interface, both materials to the left and right of the interface are now at the same shock state P_2, u_2 as shown in the pressure-distance profiles of Fig. 6b. State P_2, u_2 lies at the intersection of the reflection Hugoniot of the driver plate and the direct Hugoniot of the specimen. These conditions provide a convenient method for determining the Hugoniot of an unknown material, if the Hugoniot of the driver metal plate is known. It is known as the "impedance match method" (WALSH & CHRISTIAN, 1955). Using the impedance matching, the measurement of the shock velocity U_2 will give the magnitude of the slope $\rho_0 U_2$, where ρ_0 is the initial density of the specimen, and finally the state P_2, u_2 along the unknown Hugoniot. If P_1 is varied, several P_2, u_2 states can be determined and the Hugoniot curve can be obtained by a graphical fitting of the experimental points.

A second method for determining the peak pressure in an unknown specimen is by measuring the so-called free surface velocity ("free surface method") (RICE et al., 1958). The velocity u_{fs} given to a free surface by a shock reflection should be twice the particle velocity u (if entropy effects are neglected) because it is the sum of the particle velocity due to the shock wave and the particle velocity due to the rarefaction wave reflected back into the medium, that is

$$u \approx \frac{u_{fs}}{2} \qquad \text{(free surface approximation).}$$

This can be seen from Fig. 6a by replacing Hugoniot II by the u-axis representing the Hugoniot curve of vacuum or air, so that state P_2, u_2 will be located on the u-axis. In this case $u_2 = u_{fs}$ is equal to $2u_1$. If the shock velocity U is measured simultaneously with the free surface velocity u_{fs}, P, V-states along an unknown Hugoniot can be directly calculated using the Hugoniot equations (equations (1) and (2)).

Several experimental techniques to measure particle velocity u and shock velocity U have been applied in the past (DEAL, 1962; DUVALL & FOWLES, 1963; MCQUEEN, 1963; FOWLES et al., 1970) and the reader is referred to these articles for detailed information. All methods are generally based on measurements of the free surface motion. The shock velocity can be determined by monitoring the first motion of the surface upon arrival of the shock front. If the subsequent motion of the surface is recorded as a function of time, the free surface velocity can be measured. Because of the extremely short times involved (~ 0.1—1 μsecond) the recording system has to provide a very high time resolution (~ 0.01 μsec) as well as space resolution (~ 10—100 μ).The main recording systems use optical and electrical methods. The optical methods use a rotating-mirror smear camera in connection with flash gap or inclined mirror techniques and laser interferometer instrumentation (e.g., BARKER & HOLLENBACH, 1970). The elec-

trical methods most commonly use pin contacts (for details see DEAL, 1962; DUVALL & FOWLES, 1963; DUVALL, 1968). More direct, internal measurements have been made by using flash X-ray techniques by which the shock velocity and the density of the compressed material can be determined (SCHALL, 1950, 1958; DAPOIGNY et al., 1957). Moreover, calibrated internal pressure transducers have been applied for making direct stress measurements utilizing piezoelectric, piezosensitive, thermoelectric and polarization phenomena (DUVALL & FOWLES, 1963; DORAN & LINDE, 1966). The accuracy of pressure determination by most of these methods mentioned above is in the order of $\pm$ 2 or 3%.

3.3. Recovery Shock Experiments

The experimental techniques described so far do not yield shocked material in a state suitable and recoverable for further study of permanent effects. For the study of such residual effects it would be most desirable to recover material which has not experienced uncontrolled tension waves arising from the surfaces. Since rarefaction waves can never be eliminated, most experimental set-ups are designed to prevent rarefaction waves from forming at lateral and bottom surfaces, leaving only those arising from the shocked surface plane. Some of the recovery techniques used in the past for the investigation of geological materials do not fit these requirements, and hence, are highly unreliable in terms of pressure accuracy and of knowledge of the pressure-temperature history of the shocked specimen. This is true mainly for non-planar shock geometries as used in the implosion tube technique or other geometries producing convergent shock waves.

There are two main groups of techniques for recovery shock experimentation: plane wave geometries, and non-planar, convergent or divergent shock wave geometries.

a) Plane wave geometries

The most adequate and most widely used technique is known as the "momentum trap concept" (MILTON & DE CARLI, 1963; DORAN & LINDE, 1966). In this method the flat specimen is surrounded by a number of loose, but tightly fitting plane slabs as shown in Fig. 7. Ideally the slabs should consist of a material having shock impedance similar to that of the specimen in order to minimize wave interactions at the specimen-slab interfaces. The momentum traps protect the specimen from tension waves propagating inward from the outer surfaces upon arrival of the shock front because the slab-slab interfaces will not support tension. The slabs fly outward, leaving the specimen unaffected by tension waves and in its original position. This set-up is most adequately combined with an explosive plane wave generator as described above (Fig. 5). The final peak pressure in the specimen can be estimated with reasonable accuracy (DE CARLI & MILTON, 1965) using explosive systems with known characteristics and provided the Hugoniot curve of the specimen is known. Dr. U. HORNEMANN at the Arbeitsgruppe für Ballistische Forschung, Weil a. Rh., Germany has successfully applied electrical pin recordings for shock parameter determination in connection with the momentum trap principle in a cooperative program of shock studies on silicate minerals and glasses with the Mineralogical Institute of Tübingen (MÜLLER & HORNEMANN, 1969; ARNDT et al., 1971). In his method separate calibration runs with the same explosive and flyer plate characteristics as for the recovery run are performed. The shock pressure is then determined by graphical impedance match solutions based on measured free surface velocities of the impacted steel target (Fig. 6) and the known Hugoniots of steel and of the specimen. The accuracy is in the order of $\pm$ 4%. In this arrangement, only the uppermost part of the specimen is representative of the calculated peak pressure because it is unaffected by lateral rarefactions arising from the specimen-steel container interface. Peak pressures up to approximately 300 kbar can be achieved in most silicate minerals via a single shock transition. Peak pressures up to 600 kbar or more are accomplished in silicates by using thin discs of some 0.5 mm thickness. They result from multiple shock waves caused by reverberations at the steel-specimen interfaces (Fig. 8). The geometrical parameters of flyer plate, steel container and specimen thickness have to be arranged in a way that the final peak pressure is reached by the shock reverberations before the pressure is attenuated by rarefaction from the back surface of the flyer plate. An instructive numerical

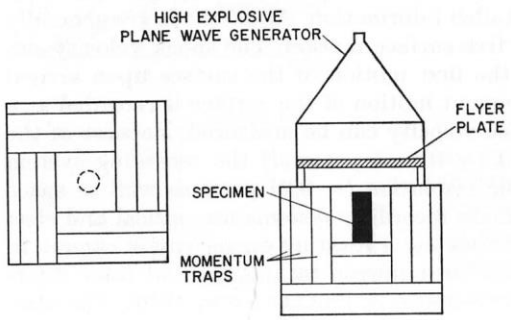


Fig. 7. Experimental set-up for shock recovery studies based on the principle of "momentum traps" (after MÜLLER & HORNEMANN, 1969).
a) side view
b) vertical view of the momentum trap assembly

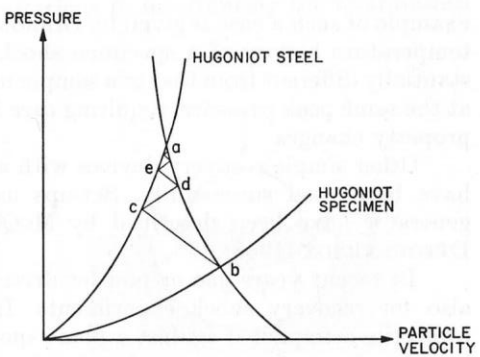


Fig. 8. Graphical representation of the "impedance matching" method in the pressure-particle velocity plane. The impact of a steel plate into a thin specimen embedded in steel produces an initial shock wave in the specimen with peak pressure b, which increases by shock reverberations (b-c-d-e-a) at the steel specimen interfaces until peak pressure a is reached.

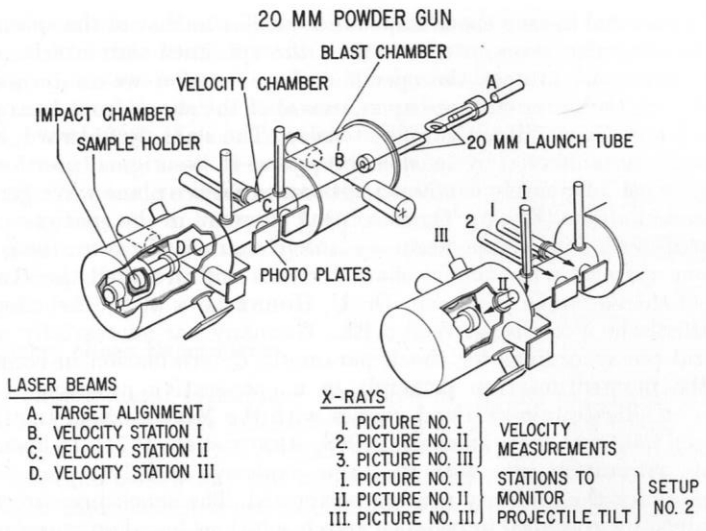


Fig. 9. Experimental set-up of a powder driven ballistic range for shock recovery experiments as used at NASA-Ames Research Center, Moffett Field, Ca. (from Hörz, 1970b).

example of such a case is given by Gibbons & Ahrens (1971). The pressure-temperature history of a specimen shock-loaded by this technique is substantially different from that of a sample undergoing a single shock transition at the same peak pressure, requiring care in the interpretation of permanent property changes.

Other simple recovery devices with and without momentum traps also have been used successfully. Set-ups using a "mouse-trap" plane wave generator have been described by McQueen (1963) and by Müller & Défourneaux (1968).

In recent years gas or powder driven ballistic ranges have been used also for recovery shock experiments. In these techniques a flat-shaped projectile is impacted against a plane specimen located at the end of a gun barrel in front of an impact chamber from which the shocked material can be collected (Fig. 9). Arrangements of this kind have been applied to geological materials by several workers and have been described by Short (1968a), Hörz (1968), Hörz & Ahrens (1969), and Gibbons & Ahrens (1971). Ballistic ranges are especially attractive for the pressure range below 100 kb, but higher pressure can be achieved by using multiple shock conditions as described above. The disadvantage over the momentum trap-explosive system is that the specimen cannot be recovered from its original position. In a ballistic device, however, more homogeneous pressures can be achieved by embedding the specimen in an appropriate impedance matching medium which is not the case for the widely used steel containers of the momentum trap method. Shock pressures in ballistic recovery devices are accurate to about $\pm$ 3%.

b) Non-planar wave geometries

Recovery devices using non-planar shock waves fall into two groups: (1) laboratory scale explosive systems, such as implosion tube techniques and experimental hypervelocity impact cratering, though the latter method is not used primarily as a recovery technique; and (2) megascale buried chemical or nuclear explosions in rocks as well as missile impacts.

Implosion tube techniques have been described by Deribas et al. (1966), Hamann (1966), Müller (1967) and Short (1968a). Cylindrical metal tubes filled with the specimen are placed in hollow metal cylinders surrounded by liquid or solid explosives. Upon detonation the hollow outer tube implodes transmitting a convergent shock wave to the specimen. Although relatively high peak pressures can be obtained, this method is severely limited to qualitative studies, because information on the pressure-temperature history of the specimen is very poor, or lacking.

The remaining experimental techniques — laboratory scale or military-type missile impact cratering and megascale surface or subsurface explosions — are all similar in that they produce divergent shock fronts propagating more or less spherically or hemispherically from the point of impact or explosion (Fig. 10). Calculation of peak pressure or pressure attenuation with increasing distance from the point of impact or explosion is complicated by the non-planar wave geometry, e.g., by the presence of a geometrical pressure attenuation during spherical wave propagation. Consequently, there are severe limitations in interpreting the relationship

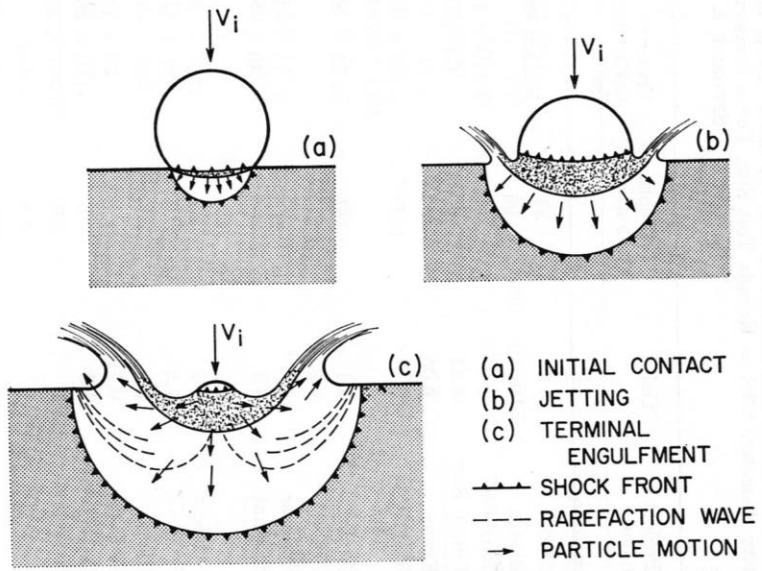


Fig. 10. Schematic representation of the shock wave propagation and particle motion at various stages of the vertical impact of a spherical body on a plane solid surface (modified after Gault et al., 1968).

Table 1. Selected surface and subsurface nuclear and TNT explosions with data on the nature of affected rocks and on post-shot investigations. Events from which substantial data on shock effects in minerals have been obtained are underlined; c = underground cavity, d = cavity diameter, NTS = Nevada Test Site. For a complete list of U.S. underground nuclear explosions during 1961—1970 see SPRINGER & KINNAMAN (1971).

Date	Project name	Yield	Depth of burial [m]	Crater or cavity dimensions [m]	Location	Rock medium	References
				Nuclear explosions			
1958	Blanca	19	96		NTS	Tuff	128, 233
1968	Cabriolet	2.3	52	110 × 37	NTS	Rhyolite	233
1962	Danny Boy	0.43	33.5	33 × 19	NTS	Basalt	125, 201, 233
1958	Evans	0.05			NTS	Tuff	128, 233
1967	Gasbuggy	29	1229	c 102 × 49	San Juan Basin, N. M.	Sandstone	233
1961	Gnome	3.1	361	c 55 × 20	Near Carlsbad, N. M.	Salt (mainly halite)	200, 233
1964	Handcar	12	402	c 71 × 42	NTS	Dolomite	233
1962	Hardhat	5.9	286	c 40 × ?	NTS	Granodiorite	29, 65, 197, 198, 200, 233
1951	Jangle-S	1.2	—1	13.5 × 6.5	NTS	Alluvium	128, 233
1951	Jangle-U	1.2	5	39 × 16	NTS	Alluvium	128, 233
1958	Logan	5.9	145		NTS	Tuff	128, 233
1958	Neptune	0.115	~ 30	30 × 10.5	NTS	Tuff	128, 233
1965	Palanquin	4.3	85	110 × 26	NTS	Rhyolite	233
1966	Piledriver	61	463	c 277 × 40	NTS	Granodiorite	29, 30, 233
1957	Rainier	1.7	274	~ 40 d	NTS	Tuff	128, 233
1969	Rulison	40	2568	c 82 × 42	Garfield County, Col.	Sandstone	233
1964	Salmon	5.3	828	35 × 35	Tatum salt dome, Miss.	Salt	233
1968	Schooner	35	108	260 × 63	NTS	Tuff	233
1962	Sedan	100	210	366 × 97.5	NTS	Granodiorite, welded tuff, orthoquartzite, limestone	35, 40, 65, 200, 202, 233
1963	Shoal	12.5	365	c 108 × 52	NTS	Granite	233
1964	Sulky	0.092	27	52 × 7	NTS	Basalt	233
1958	Tamalpais	0.07			NTS	Tuff	128, 233
1955	Teapot-S	1.2	20	44 × 27	NTS	Alluvium	128, 233
				TNT explosions			
1960	Stagecoach	0.18	24.4	35 × 2	NTS	Alluvium	163, 233
		0.18	5.2	31 × 7	NTS	Alluvium	
		0.18	10.4	35 × 9	NTS	Alluvium	
1960	Buckboard	3 × 0.18	1.5—18	(3—35) ×	NTS	Basalt	163, 233
		10 × 4.5 . 10^{-4}		(0.5—11)	NTS		
1960	Scooter	0.5	38	94 × 23	NTS	Alluvium	155, 184, 233
1964	Snowball	0.5	0	100 × 6.5	Canada	Alluvium	184

between post-shock effects and pressure-temperature history for any given particle at any position within, below or outside the crater. Nevertheless, such experimental studies are highly desirable because, unlike plane wave experimentation, they provide more appropriate models for natural shock processes where progressive shock metamorphism of the country rock is the rule. Moreover, the duration of the pressure pulse is much longer in megascale explosion events than in laboratory scale experimentation and the pressure-time profiles as well as the mode of pressure attenuation with progressive distance of propagation approach the particular characteristics of natural impact induced shock processes.

Laboratory techniques of producing craters by impacting small projectiles of various composition into flat targets have been developed and intensively applied to rock targets by D. GAULT and co-workers at NASA-Ames Research Center, Moffet Field, Ca. (CHARTERS et al., 1957; GAULT et al., 1968 and references therein). Projectiles (mostly spheres of iron, aluminum, glass, plastics) are accelerated by a light gas or powder gun to velocities of up to 8 km/sec and impacted at various angles against the target producing craters with diameters of the order of 25 cm. Since most post-impact investigations have been devoted to the crater morphology and the cratering mechanics (OBERBECK & QUAIDE, 1967; GAULT et al., 1968; QUAIDE & OBERBECK, 1968; OBERBECK, 1971) only few studies have been devoted to permanent shock effects in the ejected materials or the subcrater target material. HÖRZ (1969) studying a 30 cm impact crater in granite produced by the NASA-Ames Ballistic Range found only weak shock effects in both the ejected and in situ material. This observation agrees quite well with theoretical calculations which show that less than 10% of the ejecta experienced pressures higher than 100 kb. Consequently the rock volume affected by pressures causing unequivocal shock effects (pressure > 100 kb for silicates) is very small in this type of experiment. Apart from the short duration of pressure pulses this is a major obstacle for detailed model studies on the degree and distribution of shock damage in the impacted material which could be applied to natural large scale impact metamorphism.

These problems do not arise in megascale chemical or nuclear explosions. A number of most useful results on the formation conditions of shock effects in rock-forming minerals have indeed been obtained by the study of rock specimens from nuclear explosions performed mostly at the US Atomic Energy Commission Nevada Test Site (CUMMINGS, 1965; SHORT, 1966, 1968b, 1970a; JAMES, 1969a; BORG, 1970, 1972; Table 1) and from TNT-explosions (RODDY, 1971; Table 1). Recently, data from French nuclear explosions performed in granite of the Taourirt Tan Afella mountains (Hoggar, Sahara) have become available (e.g., FAURE & GINGLINGER, 1970, see also WEST & KELLY, 1971). It is beyond the scope of this article to describe technical details of underground nuclear explosions and the reader is referred to the literature (e.g., Geophysical Laboratory — Lawrence Radiation Laboratory Cratering Symposium, 1961; TELLER et al., 1968; Symposium on Engineering with Nuclear Explosives, 1970; WEST & KELLY, 1971). Some of the Nevada tests have been instrumented to record peak pressures and particle movements. This information has made it possible to carefully document and interpret specific shock effects in rocks collected in extensive re-entry and post-shot drilling programs (see BORG, 1972).

Missile impacts of military type like those at White Sands Test Range, USA, though intensely studied in terms of cratering mechanics (MOORE, 1971), have not been investigated for shock metamorphic effects in the target material, mainly sand, gypsum and clay.

4. Conditions of natural shock processes

Natural shock processes affecting minerals and rocks are the result of interplanetary collisions of cosmic bodies with planetary surfaces. As pointed out above this type of hypervelocity impact generates a spherical shock wave which propagates forward through the ground as well as backward through the impacting body. It attenuates very rapidly with progressive distance from the point of impact due to the spherical, non-planar propagation (Fig. 10). According to these geometrical conditions the target rocks are affected in a more or less hemi-spherical volume having a strong radial pressure gradient. Upon arrival of the shock front at the rear side of the impacting body as well as at the ground surface outside the impacted zone rarefaction waves are reflected back into the highly compressed projectile and downward into the ground leading to a high velocity upward motion and excavation of a circular crater (Fig. 10). The physical conditions of impact and excavation mechanics in rock targets have been the object of intense experimental and theoretical studies in the past decade. Since it is not the primary object of this paper to review the results of those studies the reader is referred to the literature (GILVARRY & HILL, 1956; ÖPIK, 1958; BJORK, 1961; SHOEMAKER, 1963; BALDWIN, 1963; GAULT & HEITOWIT, 1963; EICHELBERGER, 1965; GAULT et al., 1968; DAVID, 1969; BRASLAU, 1970; MOORE, 1971; OBERBECK, 1971). Though results of experimental and theoretical impact cratering studies certainly have important implications in the formation of permanent shock effects in the mineral constituents of the affected rocks, no experimental or theoretical studies have been especially devoted to the influence of impact-induced shock processes on the pressure-temperature history and consequently on the post-shock properties of minerals and rocks. This is a severe problem in the interpretation of post-shock effects induced by spherical wave geometries which imply particular rarefaction wave geometries and crater excavation processes. Consequently the excavated and in situ shocked rocks will have experienced very specific and variable pressure-temperature histories. The shock effects in these rocks have been interpreted until now almost exclusively on the basis of experimental observations or theoretical calculations in connection with plane shock wave geometries and recovery shock experiments characterized by very different rarefaction wave geometries. Especially important is the fact that spherical waves, contrary to plane waves, attentuate geometrically with propagation leading to basically different stress profiles when compared with the plane wave geometry. The amplitude of the elastic

precursor may decrease below the HEL although a second (plastic) wave front is still present. Additionally, the compression waves are followed by a tension wave (Fig. 11) which should have a significant influence on the formation of particular post-shock effects (Doran & Linde, 1966; Johnson, 1968).

Furthermore, strain rates and peak pressure duration which are distinctly different from those achieved in laboratory experiments, as well as the porosity of natural rock complexes, though relatively small in crystalline rocks, tend to lower the minimum peak pressure for the development of particular shock effects in natural impact processes when compared with the corresponding pressures found in laboratory shock experiments with single crystals. This was shown by the study of rocks from nuclear explosions (Borg, 1972).

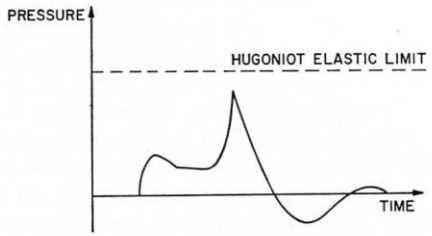


Fig. 11. Qualitative representation of the final shock wave profile after attenuation below the dynamic elastic limit for a shock wave model of spherical geometry (replotted after Doran & Linde, 1966, see text).

Since the majority of residual shock effects in rock-forming minerals have been studied in rocks and breccias of large terrestrial impact structures, the geological implications of natural impact cratering might be briefly discussed here. Based on particular shock effects, rocks at impact craters have been classified according to the principle of "progressive shock metamorphism" which follows from the attenuation of pressure and temperature with progressive distances from the point of impact as described above (Stöffler, 1965, 1966; Chao 1966, 1968; Dence, 1968; see also Fig. 10). Five main stages of metamorphism (0—IV) assigned to particular pressure-temperature ranges have been recognized and the implications of this classification are discussed in detail by Stöffler (1971a). Rocks of various stages of shock metamorphism occur in geologically different types of impact formations. They may belong either to the excavated mass ("fallout and fallback material") or to the autochthonous rock basement of a crater. Large mappable rock complexes have experienced either low shock pressure (stages 0—II) and occur as in-situ-breccias or are highly shocked (stage IV) and form thick layers of impact melt at the crater bottom or as annular rings around central uplifts. Small shocked rock fragments may be incorporated in loose detritus or in consolidated breccias consisting of a chaotic mixture of rock and mineral debris of all stages of metamorphism. These conditions may be further complicated by multiple impact, the general process on the lunar surface. The geological aspects of the various types of impact formations have been quite fully discussed by Engelhardt (1971) and Dence (1971). Only few data on the relative proportions of rocks of the various

stages of metamorphism in the ejecta blankets are known (Engelhardt et al., 1969; Abadian, 1971). Theoretical and experimental impact cratering studies indicate that on impacting solid rocks certainly less than 10%, probably only about 1% of the total ejected mass will exhibit notable shock effects (Quaide & Oberbeck, 1969; Hörz, 1969).

5. Deformation and transformation processes

Any shock wave passing through a solid may produce changes in the physical state of the solid. The changes will be either transient and reversible and have to be studied while the solid is under compression or they may be permanent and irreversible and can be studied after pressure release in the recovered specimen. It has been convenient to define residual shock effects as being produced by a shock wave with an amplitude exceeding the Hugoniot elastic limit of the material in question although irreversible changes may be caused by elastic waves. The changes are then mostly due to the action of rarefaction waves. Generally one has to be very careful in attributing observed material changes unequivocally to the actual shock compression. Most effects are achieved by the combined action of the shock wave and the accompanying rarefaction waves and the evaluation of the actual stress and temperature history of a sample recovered from an experiment or from a natural impact crater is the most serious problem of the study of residual shock effects (see e.g., de Carli, 1968).

In relation to this particular problem, only limited progress has been made in the interpretation of shock effects in rock-forming minerals in the past. The following chapter therefore will mainly deal with a more or less descriptive review of naturally or experimentally observed shock effects in minerals though some interpretations in terms of stress and temperature history will be given as far as experimental data are available. The limitations of such interpretations have been discussed above and the reader is referred to the appropriate chapters.

Available data on the behavior of rock-forming minerals under dynamic compression are summarized in Table 3.

5.1. General Aspects

The residual changes of rock-forming minerals upon shock compression may be classified most appropriately following the characteristics of the Hugoniot curve of a mineral displaying a dynamic elastic limit and a phase transition under shock compression.

With increasing peak pressure we observe the following main residual effects upon pressure release:

1. Fracturing, attributed to shock states along the region of the elastic Hugoniot (see Fig. 2b) where the formation of fractures, which may sometimes be crystallographically controlled, is due generally to tension waves following the compression. Fracturing may be caused also by yielding upon shock loading at pressures above the dynamic elastic limit.

Table 2. Frequency of the orientation of planar fractures and planar elements in naturally and experimentally shocked quartz. Frequency of a particular crystallographic form is the number of sets parallel to this form in percent of all measured sets of planar structures. This method of frequency calculation does not take into account the different numbers of symmetrically equivalent planes of the various forms and the restrictions due to inherent spatial limitations of the universal stage (see ENGELHARDT et al., 1968). n.i. = not identified.

Form	Ries Crater (crystalline rocks) ENGELHARDT & BERTSCH (1969) 300 sets per rock specimen										West Clearwater Lake (crystalline rocks) ENGELHARDT et al. (1968) 160 sets per specimen	
	B 10	B 51	S 289	B 36	B 151	B 1	S 350	S 349	B 7	B 9	Thin section	Powder mount
c (0001)	6	5	0	3	0.2	5	3	0.2	1	1	7	4
ω (10$\bar{1}$3)	75	70	44	38	42	29	34	33	38	37	35	24
π (10$\bar{1}$2)	1	2	22	22	19	17	30	25	36	27	16	18
r,z (10$\bar{1}$1)	6	7	13	12	16	15	20	14	4	14	16	17
ξ (11$\bar{2}$2)	2	4	2	4	3	4	0.3	3	3	2	3	5
s (11$\bar{2}$1)	0.5	1	4	2	2	3	1	4	3	2	5	5
m (10$\bar{1}$0)	2	2	2	3	3	5	3	4	2	4	8	13
a (11$\bar{2}$0)												
(21$\bar{3}$1)	2	3	6	8	7	8	7	6	7	8	5	7
x (51$\bar{6}$1)	4	4	5	4	4	5	2	5	4	4	4	5
(22$\bar{4}$1)	0	0	0	0	0	0	0	0	0	0	0	0
T (40$\bar{4}$1)	0	0	0	0	0	0	0	0	0	0	0	0
k (51$\bar{6}$0)	0	0	0	0	0	0	0	0	0	0	0	0
(31$\bar{4}$1)	0	0	0	0	0	0	0	0	0	0	1	1
n.i.	1.4	3	3	4	4	11	0.6	6	3	1.3	9	17
mean refractive index	1.546	1.546	1.545	—	1.536	1.534	1.533	1.529	1.480	1.479		

Table 2 (Concluded)

Form	10 Canadian meteorite craters (crystalline rocks) ROBERTSON et al. (1968) 178 sets	351 sets	Arizona Crater (sandstone) BUNCH & COHEN (1964) 1758 sets	Steinheim Basin (sandstone) BERTSCH (1968) This work		Shock experiments (single crystal quartz) WOLFE & HÖRZ (1970)	
	Type C	Type D		Sample 1	Sample 2	Cleavage	Planar features
c (0001)	8	0	29	27	41	37	31
ω (10$\bar{1}$3)	57	40	0	12	6	3	33
π (10$\bar{1}$2)	0	34	0	3	2	5	14
r,z (10$\bar{1}$1)	5	6	39	48	10	19	6
ξ (11$\bar{2}$2)	1	1.5	22	0	8	0	0
s (11$\bar{2}$1)	0	1	0.3	0	6	0	0
m (10$\bar{1}$0)	1	0	9	3	0	4	2
a (11$\bar{2}$0)							
(21$\bar{3}$1)	0	0	0	1	0	0	0
x (51$\bar{6}$1)	0	0.5	0	5	4	0	0
(22$\bar{4}$1)	14	10	0	0	0	0	0
T (40$\bar{4}$1)	0	0.5	0	0	0	0	0
k (51$\bar{6}$0)	0	0.2	0	0	0	0	0
(31$\bar{4}$1)	0	0	0	0	0	0	0
n.i.	14	7	8	?	23	10—15	

Table 3. Sources of available data on the shock wave behavior of rock-forming minerals. References are given by numbers, those in parentheses indicate data on tektites which are presumably of terrestrial origin. * Du Pont de Nemours Co., Netherland Patent Release No. 6506395.

Mineral	Chemical composition	Experimental data			Natural observations				
		Hugoniot data	Release adiabat data	Temperature data	Laboratory recovery experim.	TNT and nuclear explosions	Terrestrial impact craters	Lunar rocks	Meteorites
Alkali feldspar	$(K, Na)AlSi_3O_8$	5, 9	9	9	124, 132, 194, 217	30, 198, 200	13, 35, 38, 74, 136, 208, 210, 212, 243	100	
Amphibole	$(Ca, Na, K)_{2-3}(Mg, Fe, Al, Ti)_5(OH)_2[Si_3(Si, Al)O_{11}]_2$				124	30	50, 51, 52, 208, 210, 212		
Andalusite	Al_2SiO_5	10, 147							
Apatite	$Ca_5(OH, F, Cl)[PO_4]_2$				194		1, 38, 85, 210		
Beryl	$Be_3Al_2[Si\text{-}O_{18}]$				124				
Biotite	$K(Mg, Fe)_3(OH_2[AlSi_3O_{10}]$				119, 120, 123, 190	30, 62, 198,	35, 38, 50, 51, 52, 120, 190, 207, 208, 210		
						200	184		
Calcite	$CaCO_3$	3							
Cassiterite	SnO_2	27							
Clinopyroxene	$(Mg, Fe, Ca)(Si, Al)O_3$				45, 117, 124, 192	125, 201	127, 212, 252	57, 77, 100, 101, 175, 192, 193, 203	105, 116, 117, 138, 171, 205
Coesite, see quartz	SiO_2								
Corundum	Al_2O_3	147, 238, 244		10	124, 245				
Diamond, see graphite	C	250							
Fluorite					239				
Garnet	$(Mg, Fe, Ca)_3(Al, Fe)_2[SiO_4]_3$	8, 27			124		1, 85, 210		
Graphite	C	15, 79, 250			70, *		89, 90, 91		115, 142, 143, 165
Halite	NaCl	27				200			
Hematite	Fe_2O_3	10, 147			194				
Ilmenite	$FeTiO_3$				156		87, 89	57, 101, 156, 192, 193	
Iron	Fe	27, 151		146	18				142, 144
Jadeite	$NaAlSi_2O_6$	147							
Hex. diamond, see graphite	C								
Magnetite	Fe_3O_4	10, 147			194		87, 89		
Majorite, see orthopyroxene	$(Mg, Fe)SiO_3$								
Orthopyroxene (enstatite, hypersthene, bronzite)	$(Mg, Fe)SiO_3$	6, 10, 147, 150		7, 10	45, 124, 172, 194				170, 179
Olivine (forsterite, fayalite)	$(Mg, Fe)SiO_4$	10, 12, 147, 150		7, 10				46, 47, 57, 100, 192, 203	26, 45, 138, 162, 237
Periclase	MgO	2, 16, 147		7,10	133, 245				
Phyrrotite	FeS				194				
Plagioclase	$NaAlSi_3O_8 — CaAl_2Si_2Si_2O_8$	3, 5, 9, 10, 150	5, 9	7, 9	72, 124, 160, 194, 217	30, 125, 198, 200, 201	35, 38, 39, 40, 50, 51, 52, 85, 95, 97, 126, 183, 209, 210	57, 77, 100, 101, 175, 192, 193, 203	25, 81, 138, 225, 231
Pyrite	FeS_2								
Pyrolusite	MnO_2	10, 27							
Quartz	SiO_2	3, 5, 103, 222, 223, 227	3, 5, 169	3, 5, 7, 10	118, 124, 161, 199	30, 35, 39, 65, 198, 200, 202	35, 36, 37, 38, 40, 42, 50, 51, 52, 73, 74, 94, 96, 97, 98, 130, 183, 210, 214, 216, 221, 243, 252	77, 192, 203	(22, 48, 113, 229, 232)
Ringwoodite, see olivine	$(Mg, Fe)SiO_4$								
Rutile	TiO_2	141, 149			141, 149		87, 89		(113)
Scapolite	$Na_4Cl[AlSi_3O_8]_3 — Ca(CO_3)[Al_2Si_2O_8]_3$						85, 234		
Serpentine	$Mg_6(OH)_8Si_4O_{10}$	147							
Sillimanite	Al_2SiO_5	10, 147					211		
Sphene	$CaTiSiO_5$					29, 30	1, 52, 89, 210		
Spinel	$MgAl_2O_4$	27							
Stishovite, see quartz	SiO_2	11, 148							

2. Plastic deformations, partly combined with a partial transformation of the initial phase to a high-pressure-phase, and achieved upon release from final states along the region of the plastic Hugoniot above the HEL. If a mixed-phase region is present, this kind of deformation may be achieved upon release from all final shock states along the mixed-phase region (as far as the low pressure phase is still present) in addition to the shock states between the HEL and the onset of the mixed-phase region (see Fig. 2c).

3. Phase transformations achieved upon release from final shock states along the mixed-phase or high-pressure-phase Hugoniot (see Fig. 2c) where the material is partially or completely transformed to a high pressure phase under shock compression. Upon pressure release the high-pressure-phase may have reverted partly to the low-pressure-phase or to a short-range-order phase.

4. Thermal dissociation or melting achieved upon release from shock states along the transformational Hugoniot leading to very high shock and post-shock temperatures which may even exceed the vaporization temperatures of the shocked material.

Not all minerals show this complete sequence of possible residual shock effects and the above classification has to be considered only as a general outline. There are several factors governing the type and number of particular shock effects achievable in a mineral over a certain range of shock pressures. The following main points have to be considered:

a. Shape of the Hugoniot curve
b. Pressure-time and temperature-time profile of the whole shock transition
c. Crystallographic orientation with respect to the shock front within the low pressure phase region of the Hugoniot.

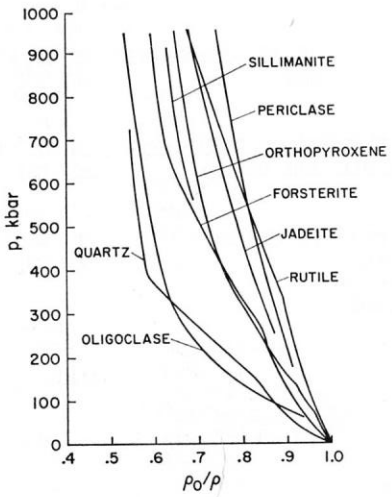


Fig. 12. Hugoniot curves of various rock-forming minerals. Data are from the following sources: quartz (WACKERLE, 1962), oligoclase (AHRENS et al., 1969a), forsterite (AHRENS et al., 1970b; MCQUEEN & MARSH, 1966), orthopyroxene (AHRENS & GAFFNEY, 1971; MCQUEEN & MARSH, 1966), sillimanite (MCQUEEN & MARSH, 1966), jadeite (MCQUEEN & MARSH, 1966), periclase (MCQUEEN & MARSH, 1966).

The most important factor is the shape of the Hugoniot curve of the mineral in question which expresses its dynamic compressibility. This property is directly dependent on the mineral's crystal structure or packing density and its capability of undergoing polymorphic phase transitions upon compression. This may be illustrated by the Hugoniot curves of some rock-forming minerals as plotted in Fig. 12. Minerals with low packing densities (low symmetry of the crystal lattice) having open structures, the framework silicates for example, display a distinctly greater compressibility than more densely packed structures like those of pyroxene, olivine or periclase and are characterized by marked mixed-phase regions with abnormally large compressibilities. Consequently, we observe distinctly different types of post-shock effects in those minerals at a given peak pressure. Consider the common case of coexisting mineral phases in a polycrystalline rock: one phase may have undergone a complete phase transition (e.g., plagioclase) whereas another one may only display a particular plastic deformation (e.g., pyroxene), neglecting at this point the complications of the shock propagation in polycrystalline, multiphase rocks (see below).

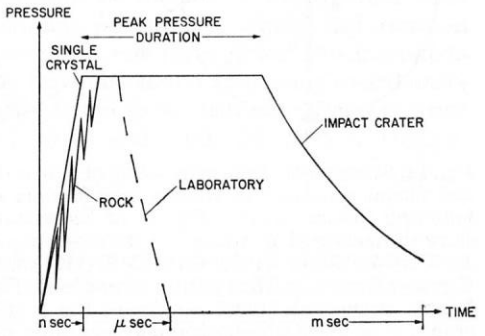


Fig. 13. Schematic graph of the pressure-time-characteristics of various shock processes (modified after KIEFFER, 1971; see text).

The second factor, the pressure-time and temperature-time profile of a shock transition, reflects mainly the type of physical process generating a shock front and its geometry of propagation (see Fig. 1 b, c, d) as well as some physical properties of the shocked material such as yielding, phase transitions, and viscosity. These properties determine the stability of the shock front (single or multiple shock waves) as well as possible stress relaxation effects. Most conspicuous are the differences in the duration of the pressure pulse and strain rates for various shock processes. Laboratory scale experiments have pulses in the order of 10^{-7}—10^{-5} sec whereas this time may increase up to 10^{-2} to 10^{-1} sec or more at megascale explosions or natural impacts of large meteoroids (Fig. 13). This fact has a particularly important influence on reactions taking place during compression such as the formation of crystalline high pressure phases. For example, it is well known that coesite is not formed in dense quartzose materials by laboratory scale experiments with short pressure pulses (DE CARLI & MILTON, 1965). The influence of pressure duration on certain residual effects has not been studied quanti-

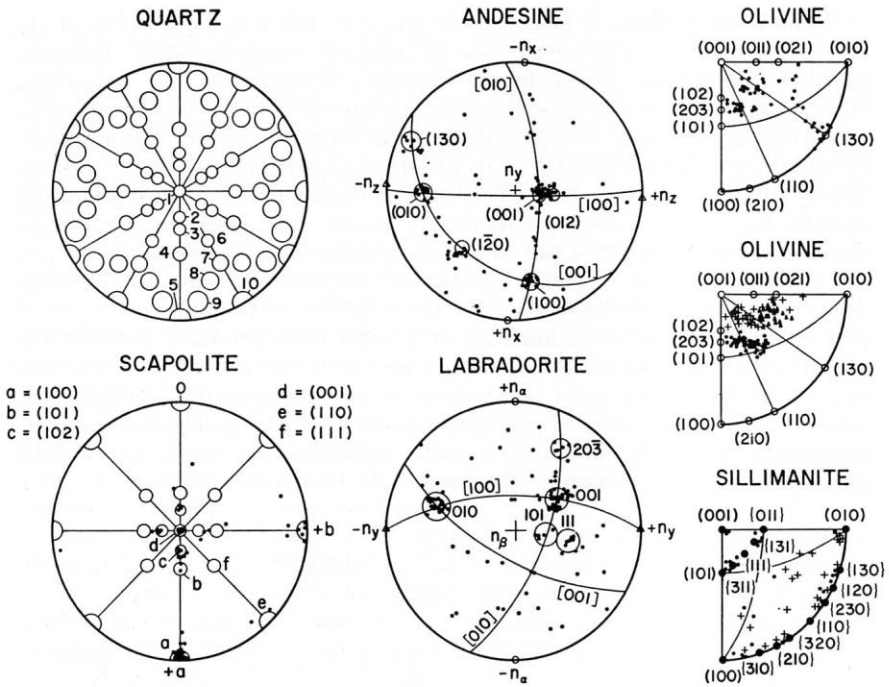


Fig. 14. Stereoplots of the orientation of planar deformation structures (planar fractures and planar elements) in various rock-forming minerals. Data are replotted from the following sources: *quartz:* Fig. 10 of ENGELHARDT & BERTSCH (1969). The numbers show the measured orientation of the most frequent planar elements: 1. (0001) 2. (10$\bar{1}$3) 3. (10$\bar{1}$2) 4. (10$\bar{1}$1) 5. (10$\bar{1}$0) 6. (11$\bar{2}$2) 7. (11$\bar{2}$1) 8. (21$\bar{3}$1) 9. (51$\bar{6}$1) 10. (11$\bar{2}$0). — *scapolite:* Fig. 4 of WOLFE & HÖRZ (1970). — *andesine:* Fig. 3 of STÖFFLER (1967). — *labradorite:* Fig. 11 of DWORAK (1969). — *olivine:* Fig. 7 of MÜLLER & HORNEMANN (1969); upper graph: experimentally shocked polycrystalline olivine (poles of planar elements parallel to pinacoids are not plotted); lower graph: experimentally shocked single crystals (dots, triangles and crosses correspond to shock propagation directions [100], [010], and [001], respectively). — *sillimanite:* Fig. 2 of STÖFFLER (1970a); points: planar fractures crosses: planar elements.

tatively. Some plastic deformations such as microscopic planar deformation structures seem not to be influenced in their mode of occurrence and frequency distribution of their crystallographic orientation by even large variations of the pressure duration as shown by a comparison of recovery shock experiments (e.g., MÜLLER & DÉFOURNEAUX, 1968; HÖRZ, 1968) with naturally shocked minerals from large impact craters (e.g., ENGELHARDT & BERTSCH, 1969). Little is known about the effect of the remaining parameters such as rise and decay time of pressure and temperature. These vary considerably among the various types of experimental and natural shock processes (Fig. 13). Additionally, since most natural materials are polycrystalline, multiphase rocks, the pressure/temperature-time profile achieved in a particular mineral may vary locally from stress wave interactions at phase or grain boundaries and free surfaces which cause local stress variations in the microscale. Equilibrium shock pressure will be reached via shock reverberations thereby extending the pressure rise time as well as decreasing the irreversible heat. Hence particular crystals of polycrystalline aggregates will have individual pressure and temperature histories depending on their textural relationships to the adjacent crystals and on differences in their physical properties. The implications of this fact for the interpretation of shock effects in particular rock constituents will not be discussed in detail here, but the reader can get more insight into this problem from an article by KIEFFER (1971) which deals with the shock wave propagation and energy dissipation in porous sandstones.

A third factor considered here is the ralationship between the crystallographic orientation of a crystal with respect to the shock front and its mode of deformation. This refers only to shock compression in the low pressure phase region of the Hugoniot where a material's rigidity and the anisotropic nature of crystals cannot be neglected. Experimental observations on quartz (HÖRZ, 1968) and olivine (MÜLLER & HORNEMANN, 1969) reveal a strong dependence of the orientation and frequency of plane microdeformation structures on the direction of the shock propagation with respect to the crystal's crystallographic orientation. Planar deformation structures parallel to (0001) in quartz do not form if the shock front is parallel to (0001) (HÖRZ, 1968). The orientation and frequency distribution of planar deformation structures were found to be distinctly different after shocking olivines parallel to [001], [010] and [100] (MÜLLER & HORNEMANN, 1969; Fig. 14).

5.2. Fracturing

Irregular fractures are commonly observed in all shocked minerals over a wide range of peak pressures and are therefore, not indicative of a particular pressure. However, at peak pressures below the material's HEL fracturing may be the only conspicuous residual effect indicating a preceding dynamic compression though it is not unequivocal. The intensity of fracturing has been observed to be proportional to the peak pressure. In correlating peak pressure and fracturing intensity a fracturing index has been used by measuring the number of fractures per sectioned area in microscopical thin section of shocked rocks and minerals (SHORT, 1965). Thereby the fracturing index of granitic rock samples was successfully correlated with the calculated shock pressures at nuclear explosion craters (SHORT, 1966, 1968b; BORG, 1972) and laboratory impact craters (HÖRZ, 1969).

5.3. Plastic Deformation

This section considers all types of shock effects caused by plastic deformation at shock pressures generally exceeding the measured material's HEL which is in the order of 20—120 kbar for most silicates[1]). Aside

[1]) Dr. I. Y. BORG (personal communication) pointed out that these pressure limits might not represent the true elastic limits. This means that also irregular fracturing occurring at pressures as low as about 6—8 kbar (BORG, 1972) may be mainly due to yielding upon shock loading rather then to tensional stresses upon pressure release.

from irregular fractures formed on shock loading we might distinguish between

a) planar fractures
b) planar elements (planar features, shock lamellae)
c) deformation bands
d) irregular plastic lattice deformation (mosaicism)

a) Planar fractures

Planar fractures have been observed frequently in quartz, olivine, sillimanite, scapolite, garnet, apatite, periclase and rarely in feldspars at terrestrial impact structures, nuclear explosion craters and laboratory shock experiments.

Planar fractures recognized as open fissures occurring in parallel sets (Fig. 15) are generally parallel to crystallographic planes of low Miller indices

(BUNCH & COHEN, 1964; ENGELHARDT & BERTSCH, 1969; MÜLLER & HORNEMANN, 1969; STÖFFLER, 1970a). These fractures have been observed in some statically deformed minerals but show different frequency distribution. Their dynamic origin may be doubtful in some cases unless additional characteristic shock effects are present in the same mineral such as planar elements or partial isotropization. Planar fractures are closely related to planar elements described below and it may be difficult or impossible to make a clear distinction between these types of deformation structures. However, they form a genetically unique group because their formation often precedes the formation of planar elements for which planar fractures sometimes act as grain boundaries as has been observed in quartz and sillimanite (ENGELHARDT & BERTSCH, 1969; STÖFFLER, 1970a). It can be inferred from shock recovery experiments on quartz (MÜLLER & DÉFOUR-

Table 4. Crystallographic orientation of planar deformation structures in shocked minerals. Formation pressures for the various types of microstructures are listed. (1) = some forms rarely occur as planar fractures, (2) = some forms occur also as planar fractures, * observed only by ROBERTSON, et al. (1968); p.f. = planar fractures, p.e. = planar elements, m.t. = mechanical twinning.

Planar deformation structures	Quartz	Andesine[1]	Labradorite	Scapolite	Pyroxene	Amphibole
Planar fractures	(0001) {10$\bar{1}$1} $\vee$ 50 kbar $\vee$				(001) (010) {h01} {hkl} $\gtrsim$ 350 kbar	(100) (001) {110} {hk0} {h0l} {hkl} $\gtrsim$ 350 kbar
Planar elements	{10$\bar{1}$3} {10$\bar{1}$2} {10$\bar{1}$1} {11$\bar{2}$2} {11$\bar{2}$1} {10$\bar{1}$0} {11$\bar{2}$0} {21$\bar{3}$1} {51$\bar{6}$1} {22$\bar{4}$1} {40$\bar{4}$1} {51$\bar{6}$0} {31$\bar{4}$1} 120—380 kbar	(001) (010) (100) (1$\bar{2}$0) (012) (130) (101) (111) (20$\bar{3}$) (201) ($\bar{1}$01) (131)* ($\bar{1}$02)* (1$\bar{2}$1)* $\gtrsim$ 150—300	(001) (010) (111) (20$\bar{3}$) (101) (100) (012) (120)* $\gtrsim$ 150—300	{100} {110} (001) {101} {102} {103} {112} {111} {113} observed at ~ 350—450 kbar	{h0l} $\vee$ $\gtrsim$ 350 kbar	{hkl} $\vee$ $\gtrsim$ 450 kbar
Mechanical twinning					(001) and (hkl) sub-parallel to (001) $\gtrsim$ 50—450 kbar	(001) and (hkl) sub-parallel to (001) $\gtrsim$ 40—450 kbar
References	37, 96, 118, 161, 183	183, 209	85, 183	85, 234	p.f.; p.e.: 212 m.t.: 101, 212	p.f.; p.e.: 212 m.t.: 29, 212

Biotite	Olivine[2]	Sillimanite	Sphene	Garnet	Apatite	Ilmenite
(11$\bar{2}$) (111) (112) (1$\bar{1}$1) $\vee$ $\gtrsim$ 450 kbar	{100} {010} {001} {130} $\gtrsim$ {112} $\gtrsim$ {122} $\gtrsim$ {111} $\gtrsim$ {131} $\gtrsim$ 50—500 (?) kbar	{311} {131} and other bipyramids, prisms obs. at ~ 400—500 kbar		{110} {211} $\vee$ $\gtrsim$ 200 (?) kbar	(0001)	
		(010) other prisms, bipyramids observed at ~ 400—500 kbar			{10$\bar{1}$1} {10$\bar{1}$2} {11$\bar{2}$1} {11$\bar{2}$2} {31$\bar{4}$2} $\vee$ $\gtrsim$ 200(?) kbar	
			~ {221} $\vee$ $\gtrsim$ 10 kbar			{10$\bar{1}$1} (0001) $\vee$ $\gtrsim$ 50 kbar (?)
190, 212	125, 162, 191	211	29	1, 210	1, 38, 194, 210	156

NEAUX, 1968; HÖRZ, 1968) and olivine (MÜLLER & HORNEMANN, 1969) and from nuclear explosions in granite (BORG, 1972) that planar fractures can be formed by lower peak pressures than planar elements, e.g., as low as ca. 50 kbar in the case of quartz and olivine. They might be interpreted as microscopic traces of a stress relaxation process behind the elastic percursor wave (MÜLLER & HORNEMANN, 1969). If the actual elastic limit is below 10 kbar, they might result in part from yielding upon compression and in part from tensile stresses upon pressure release (BORG, 1972). Data on the orientation and formation conditions of planar fractures in various minerals are given in Table 4, Fig. 14 and Fig. 16.

b) Planar elements

These planar deformation structures occur as multiple sets of parallel optical discontinuities, sometimes resolvable as fine microscopic lamellae, with a spacing of about 2 to 10 μm in quartz, cristobalite, tridymite, plagioclase, alkali-feldspar, scapolite, olivine, sillimanite, pyroxene, amphibole, biotite, apatite, periclase found at lunar and terrestrial impact craters and nuclear explosions (Table 3). They have also been produced in most of the above minerals by laboratory and nuclear explosion experiments (SHORT, 1966; MÜLLER & DÉFOURNEAUX, 1968; HÖRZ, 1968; MÜLLER & HORNEMANN 1969; SCLAR, 1969; SCLAR & USSELMANN, 1970; BORG, 1972).

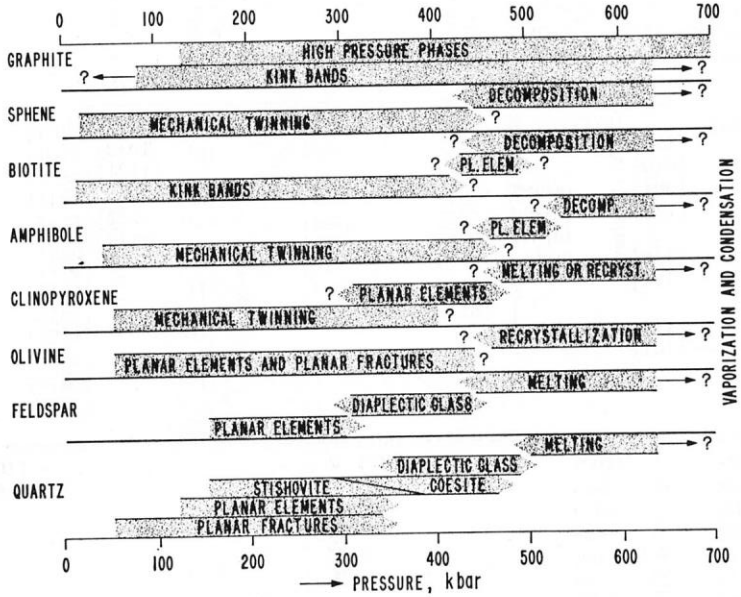


Fig. 16. Schematic representation of the pressure ranges in which various types of residual shock effects are produced in the most frequent rock-forming minerals. Question marks indicate uncertain pressure limits. The diagram is based on experimental data and natural observations which are referenced in Table 2.

Planar elements also called "planar features" (CARTER, 1965) or "shock lamellae" (CHAO, 1967a, b, 1968) have been studied most intensely in shocked quartz (see articles in FRENCH & SHORT, 1968 as well as CARTER, 1965; ENGELHARDT & BERTSCH, 1969 and references therein). Planar elements in quartz from the Ries crater have been categorized by ENGELHARDT & BERTSCH (1969) into a) nondecorated planar elements which are fine optical discontinuities (dark straight lines) not resolvable by the microscope (Fig. 17) b) homogeneous lamellae which differ slightly from the host crystal by their optical orientation and by decreased refractive index, and c) filled lamellae (Fig. 17) consisting of aggregates of minute crystals of high refractivity, presumably stishovite (STÖFFLER, 1971b). Type a) and b) elements are also the most common types of microstructures in the other rock-forming minerals mentioned above (Fig. 18—21). An additional type of planar elements is called "decorated planar elements" (ROBERTSON et al., 1968). They are to be seen microscopically as planar arrangements of minute, spherical to elliptical inclusions which may be vesicles or holes filled by gas or liquid (Fig. 22). They are very common in quartz but are also present in feldspar and amphibole. They have been found only at terrestrial impact craters where they appear to have formed by secondary thermal annealing of shocked minerals with type a)—c) microstructures caused by burial within hot layers of impact breccias.

Planar elements display a characteristic crystallographic orientation preferentially parallel to densely packed lattice planes which contain the shortest Bravais vectors (Fig. 14). Some crystallographic implications of planar elements in quartz have been discussed in detail by ENGELHARDT & BERTSCH (1969). Single grains typically are cut by two, three or more sets of microstructures (quartz has up to 15 sets) of different orientations. They are well distinguishable from "deformation lamellae" produced in some minerals by static deformation by their multiplicity and crystallographic orientation as well as by their optical properties (CARTER, 1971). The basic process of their formation appears to be crystal gliding caused by the extremely high strain rates and shear forces associated with shock transitions. Particularly three possible main processes of formation have to be considered:

a) gliding due to stress relaxation bedind the elastic wave at final shock states below the HEL
b) gliding due to yielding at final shock states above the HEL
c) reversion of a high pressure polymorph into a short-range-order phase upon release from shock states of the mixed-phase region. Gliding is presumably involved in the formation of the high pressure phase during compression.

The latter process (c) is predominant in the formation of planar elements in quartz and feldspar where the presence of a short-range-order phase has been confirmed by microscopic (STÖFFLER, 1967) and electron microscopic investigation (MÜLLER, 1969). Stress relaxation behind elastic shocks have been considered as a main cause for the formation of planar microstructures in olivine (MÜLLER & HORNEMANN, 1969).

Most detailed data on the peak pressures at which planar elements form are available from well-defined shock recovery experiments on quartz (Müller & Défourneaux, 1968 and Hörz, 1968) and on olivine (Müller & Hornemann, 1969) and from nuclear explosions in granite (Borg, 1972). Planar elements in single crystal quartz parallel to $\{10\bar{1}3\}$ first appear at pressures above ca. 120 kbar. Planar elements parallel to $\{10\bar{1}2\}$ are formed in the 160—200 kbar range and become more frequent than the $\{10\bar{1}3\}$-structures at pressures exceeding ca. 280 kbar. Borg (1972) observed a few quartz grains with weakly developed planar elements in granite at equilibrium pressures of as low as 75 kbar whereas at 155 kbar 89% of the grains contained one or more sets of planar elements. The number of multiple sets and their density generally increases with increasing peak pressure. Less frequent orientations are given in Table 2 together with the available data on the relative frequency of the particular orientations of planar elements in shocked quartz of various origin. Partial isotropization has been observed above 300 kbar and is complete at about 380 kbar.

The whole pressure range in which planar elements are formed in single crystals, that is between about 120 and 350 kbar, as observed from recovery experiments, fits closely the pressure range of the two-phase region of the Hugoniot of quartz (Fig. 17) as given by Ahrens & Rosenberg (1968). It is interesting to note that Robertson et al. (1968) were able to establish four types (A—D) of combinations of differently oriented planar elements in a suite of progressively shock metamorphosed autochthonous quartz-bearing rocks beneath the Brent crater, Canda (Dence, 1968). This progressive series A—D shows the same sequence of development of the main microstructures such as (0001), $\{10\bar{1}1\}$, $\{10\bar{1}3\}$, $\{10\bar{1}2\}$ with increasing shock pressure as observed in single crystal recovery experiments.

According to de Carli et al. (1967) and to Müller & Hornemann (private communication) the formation of planar elements in plagioclase requires pressures between about 150 and about 300 kbar, which again is in good agreement with the pressure limits of the mixed-phase region of the Hugoniot of plagioclase as given by Ahrens et al. (1969a).

Data on the orientation and on the required, estimated or measured peak pressures of planar elements is given in Table 4 and Figs. 14 and 16 for various rock-forming minerals.

c) Deformation bands

The term deformation band shall be used here in a broad sense that means for all linear, lense- to lamellar-shaped regions in a crystal differing in crystallographic orientation from the host. Thus "kink bands" and "mechanical twins" are included and distinction from undulatory extinction is made according to the definition of Carter et al. (1964). For the present purpose we may distinguish between three main types of deformation bands:

1. kink bands
2. mechanical twins
3. deformation bands with diffuse boundaries (extinction bands).

Deformation bands are generally formed by gliding and lattice rotation resulting from plastic deformation. In contrast to the planar elements their physical properties do not differ essentially from those of the host crystal. They often occur in multiple sets having individual thicknesses of some micromillimeter to some decimillimeter and are more rarely defined in terms of particular crystallographic orientations than are planar deformation structures. Since deformation bands produced by experimental or natural static deformation are widespread features in rock-forming minerals (Borg & Handin, 1966; Carter, 1971) and are very similar to their shock-induced counterparts, they are not unequivocally indicative of dynamic deformation unless they are combined with other undoubted shock effects.

1. Kink bands are most characteristic shock effects in sheet silicates and other sheet-like structures such as graphite although they may occur in shocked quartz and feldspar (Bunch, 1968). They have been observed in shocked biotite from meteorite craters (Stöffler, 1966; Chao, 1967a; Bunch, 1968) and most intensely studied by Schneider (1971). They are also known from laboratory recovery experiments and megascale explosions (Cummings, 1965; Short, 1966, 1968a, b; Hörz, 1969, 1970a; Hörz & Ahrens, 1969; Borg, 1972). Kinking in biotite is achieved by crystal gliding with the base (0001) acting as the gliding plane combined with an external rotation of the crystal lattice (Fig. 23). The same holds presumably for graphite (Fig. 24). Kink bands in biotite may be formed by dynamic pressures between 10 and about 450 kbar (Hörz & Ahrens, 1969; Schneider, 1971, Fig. 16). They are also produced by experimental and natural static deformation (Turner, 1964; Borg & Handin, 1966). However, Hörz (1970a) and Schneider (1971) have found some characteristic differences between "static" and "dynamic" kink bands in terms of their geometrical relationships to the host crystal (Fig. 23). Schneider, in a detailed study of biotite from all stages of shock metamorphism of the Ries crater, observed distinctly higher asymmetry of kink bands (higher values for δ—ε, Fig. 23) than reported from the static counterparts. The asymmetry increases with increasing shock pressure as does the intensity of kinking whereas the kink band width decreases. Apparently the gliding directions are quite variable in contrast to the static deformation where [100] is the predominant direction. This may be related to an interesting observation by Cummings (1965) that kink bands in randomly oriented biotites of a granite at the Hardhat nuclear explosion had a preferred orientation with respect to the direction of shock wave propagation: most kink bands were found at 90° to the normal of the shock front and other preferred but less frequent orientations are at 70° and 50° to the shock direction (see also Hörz, 1970a). Thus [100] can not be the only possible gliding direction.

2. Mechanical twins. Polysynthetic mechanical twinning is a very characteristic and major shock effect in the chain silicates pyroxene and amphibole as well as in sphene and ilmenite, whereas it is of minor frequency in plagioclase. The twins appear in the microscope predominantly as sets of narrow, parallel to slightly lense-shaped bands, submicroscopic to some 10 μm in width. They have been observed in terrestrial (Stöffler, 1970b; James, 1970) and lunar shock metamorphosed rocks (Carter et al., 1970,

1971; Chao et al., 1970; Dence et al., 1970; Engelhardt et al., 1970, 1971; Minkin & Chao, 1971; Quaide & Bunch, 1970; Sclar, 1970, 1971; Short, 1970b) as well as in meteorites (Hornemann & Müller, 1971; Levi-Donati, 1971). They are also present in rocks and single crystals recovered from nuclear explosions (James, 1969a; Borg, 1970, 1972) and laboratory experiments (Sclar, 1970; Hornemann & Müller, 1971). Though most investigators consider these lamellae as mechanical twins, the actual twin geometry has seldom been confirmed by X-rays, electron microscopy or universal stage optics. However, the twin origin was actually proven for ilmenite (Minkin & Chao, 1971), sphene (Borg, 1970) and diopside (Hornemann & Müller, 1971). Shock induced twinning in sphene has been studied in detail. The gliding plane is near {221} and [110] is the gliding direction (Borg, 1970). This type of twinning requires the breaking of either Si—O, Ti—O and Ca—O bonds which might explain the fact that twinning is more difficult to produce in static deformation.

In ilmenite the twin planes have been found to be parallel to $\{10\bar{1}1\}$ and (0001) (Minkin & Chao, 1971). In pyroxene twinning ist mostly parallel or subparallel to (001), rarely to (100) (e.g., Engelhardt et al., 1971, Fig. 25) and was observed in a pressure range from about 50 to 400 kb by Hornemann & Müller (1971). The same crystallographic orientation of polysynthetic twins has been measured in hornblende of the Ries crater (Stöffler, 1970b, Fig. 26) and of the Piledriver nuclear explosion (Borg, 1972), where twinning occurs at peak pressures of as low as 25—40 kbar. The twin lamellae in both clinopyroxene and amphibole have an average thickness of 2—10 μm and a spacing of 2—20 μm. Data on the orientation and formation pressures for twinning in shocked minerals is summarized in Table 4 and Fig. 16.

Mechanical twins have also been produced by natural and experimental static deformation in the described minerals. Statically formed twins generally differ in crystallographic orientation from their dynamic counterparts though some of them follow the same twin laws. "Static twins" were found parallel to $\{10\bar{1}2\}$ and $\{10\bar{1}1\}$ in ilmenite (Minkin & Chao, 1971), on {221} in sphene (Borg & Heard, 1971), on (100) and (001) in clinopyroxene (Raleigh & Talbot, 1967) and on (001) in amphibole (Rooney et al., 1970, Buck, 1970) where its cell axes are chosen according to the I 2/m lattice.

3. Deformation bands with diffuse boundaries have been reported from naturally and artificially shocked silicate minerals. They are probably most frequent in framework silicates (Bunch, 1968; Müller & Défourneaux, 1968; Stöffler, 1967) and olivine (Carter et al., 1968; Short, 1968a). These bands called "extinction bands" by Carter et al. (1968) appear as somewhat irregular, mostly lenticular shaped bands, some centimillimeter wide. Extinction bands in feldspars are subparallel to (010) (Bunch, 1968) and near $\{10\bar{1}1\}$ in quartz (Müller & Défourneaux, 1968). My observations indicate that the band orientation is greatly variable. Very little is known about the geometry of slip mechanisms and the peak pressures necessary for the production of extinction bands. They problaby form over a wide range of dynamic pressures which might be in the order of 50 to 350 kbar for most silicates.

d) Mosaicism

Mosaicism represents the most common result of plastic deformation by dynamic compression. Mosaicism, most conspicuous in microscopic observation by a highly irregular optical extinction pattern, is achieved by the formation of mutually disoriented domains or blocks in the crystal structure. This "internal fragmentation" of the crystal structure becomes most distinct in all rock-forming minerals by shock compression above the Hugoniot elastic limit although it might be accomplished by elastic waves if the material yields as a result of stress relaxation effects as well as by the action of tension waves.

Mosaicism is generally accompanied in many minerals by indications of plastic deformation such as planar deformation structures or deformation bands. It remains characteristic over a wide range of peak pressures up to the onset of a mineral's transformational Hugoniot that is the Hugoniot of the high-pressure-phase.

Mosaicism is revealed by the crystals' microscale extinction behavior under the polarizing microscope and might be characterized quantitatively to some extent by measuring the scatter of the optic exes throughout different regions of a crystal (Dachille et al., 1968). A more exact method of investigation is by X-ray diffraction of deformed single crystals. This method was used first for the study of the plastic deformation of shocked rock-forming minerals by Dachille et al. (1964) and Lipschutz & Jaeger (1966). They measured the degree of asterism in single-crystal X-ray patterns, a method long before successfully applied to the study of the plastic deformation of metals induced by static compression. Asterism is defined as the angle within which the normals of a particular lattice plane vary throughout a deformed single crystal, and is normally measured by Laue techniques. As a general rule, the shape of a diffraction spot reveals the degree of misorientation of crystal domains; spot or streak breadth is related to domain size, the broader the spot or streak the smaller the domain size.

Because of difficulties in orienting shocked single crystals as necessary for the Laue technique, most investigators have used stationary and rotational Debye-Scherrer techniques with randomly oriented crystals. However, the factors causing streakiness and spot or line broadening in Debye-Scherrer patterns are complex and the evaluation of the relationships of the pattern to the shock-induced mosaicism is complicated and not well understood as discussed by Hörz & Quaide (1972). The main phenomena revealed in such X-ray diffraction patterns are:

a) Elongation of diffraction spots at a constant diffraction angle (streakiness, not identical with Laue asterism, see Hörz & Quaide, 1972)
b) spot or line broadening
c) preferred orientation of the crystal domains

In the past, several authors have tried to correlate these phenomena with the shock pressure experienced by the crystal in question (Lipschutz & Jaeger, 1966; Dachille et al., 1968; Hörz & Quaide, 1972). The degree of streakiness (leading finally to actual diffraction rings) and spot or line

broadening generally increase with increasing peak pressure whereas "preferred orientation" gradually decreases turning to random orientation, i.e., shock-induced polycrystallinity, at pressures near the breakdown of the material either by a phase transition or by thermal decomposition. At this point mosaicism might interfere with a beginning recrystallization achieved in the shocked mineral by high residual temperatures. Both processes may lead to typical powder X-ray patterns. Consequently, any correlation between pressure and degree of mosaicism is only valid if recrystallization can be excluded. This is often difficult because shocked minerals are commonly subjected to secondary thermal metamorphism.

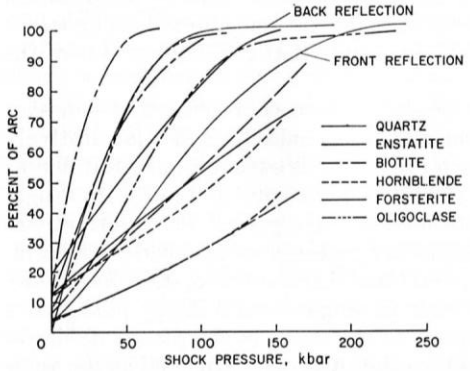


Fig. 27. X-ray diffraction streakiness ("asterism") versus peak pressure for the front and back reflections in single crystal "rotational" Debye-Scherrer patterns of experimentally shocked rock-forming minerals (replotted after Hörz & Quaide, 1972, see also text).

Based on "rotational" Debye-Scherrer X-ray patterns of experimentally shocked minerals, Hörz & Quaide (1972) established a correlation between streakiness, quantitatively measured as the percentage of a whole diffraction ring at constant diffraction angle (for details see Hörz & Quaide, 1972), and the peak shock pressure for some common rock-forming minerals (Fig. 27). With increasing pressure streakiness and finally Debye rings appear first in the back reflections indicating a decrease or loss of long range order. This occurs for the front reflections at somewhat higher pressures. Simultaneously the lines or streaks broaden for both types of reflections in the same sequence with increasing pressure (Fig. 28). Both effects are kind of "bulk" effects caused by the combined action of several mechanisms in the crystal structure such as formation of a certain domain size and internal strain. This has to be kept in mind for any quantitative calibration of these effects in terms of shock pressure especially if applied to the more complicated shock history of minerals within polycrystalline, multiphase rocks. Nevertheless, Hörz & Quaide (1972) found a characteristic sequence of resistance to shock-induced mosaicism in the investigated minerals. The stability increases in the following order: sheet silicates (biotite), framework silicates (feldspar < quartz), chain silicates (pyroxenes, hornblende), orthosilicates (olivine). This sequence roughly corresponds to a decreasing dynamic compressibility (compare Fig. 12). Dachille et al. (1968) established a similar

correlation between peak pressure and mosaicism as measured by the ratio asterism (length of streaks): line breadth for quartz of the Hardhat nuclear explosion. These authors also studied minerals subjected to static deformation up to some 150 kbar resulting in similar effects of mosaicism. Although there is some overlap in the degree of mosaicism with static deformation, shock-induced mosaicism can be distinguished easily, at least for pressures exceeding 50—100 kbar. This is especially true for tectonically deformed minerals which experienced much less pressure and extremely low strain rates.

Mosaicism, as revealed by X-ray patterns, has been widely observed in minerals of shocked rocks from impact processes on the Earth and Moon (Chao, 1968; Dachille et al., 1968; Quaide & Bunch, 1970; Stöffler, 1970a, b; Schneider, 1971). Bunch et al. (1968) and Chao (1968) found low intensity and partial or total omissions of back reflection lines in shocked quartz having low density and refractive index. Chao (1968) observed fading out of back reflections if the mean refractive index was below 1.532, but no obvious relation between X-ray asterism and refractive index of highly shocked quartz (probably > 200 kbar) was found. Quaide & Bunch (1970) used a correlation of mosaicism and shock pressure for plagioclase calibrated by recovery shock experiments for the quantitative determination of the proportion of shocked plagioclase grains in lunar

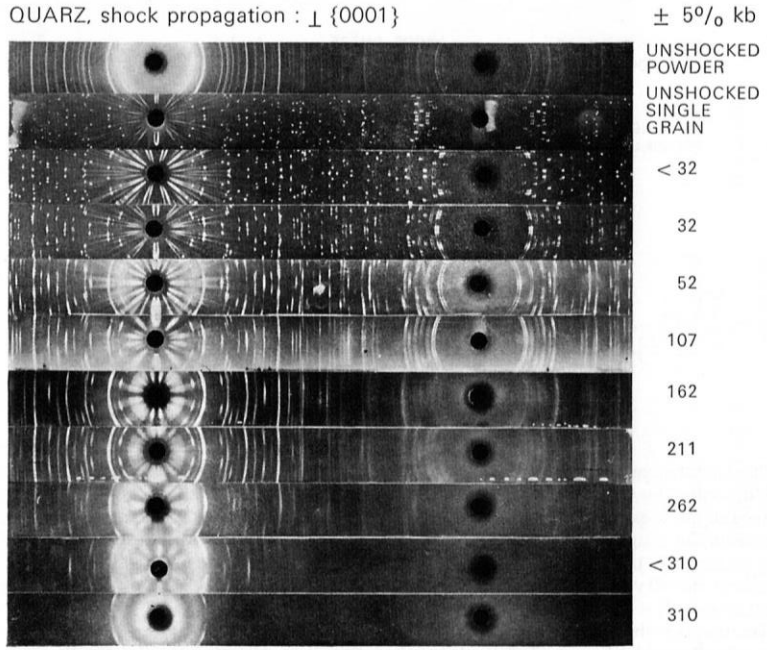


Fig. 28. Debye-Scherrer diffraction patterns of experimentally shocked quartz (photo courtesy of Dr. F. Hörz, from Hörz & Quaide, 1972).

Apollo 11 soils and breccias. Notwithstanding the problems of an application of laboratory measurements to the complicated shock history of the lunar regolith, about 50% of the plagioclase grains have experienced pressure greater than 40 kbar, 20% greater than 90 kbar, and 10% greater than 150 kbar. SCHNEIDER (1971) verified a selective mode of disordering (in shocked biotite of the Ries crater) by observing distinct differences in the degree of asterism of particular interferences (lattice planes) probably achieved by various types of gliding parallel to (001), (100) ($\bar{1}32$) and (010).

5.4. Solid State Transformations

Solid state transformations include two types of transitions: (a) phase transitions to high-pressure-phases and (b) transitions to low density, short-range-order phases of the same composition. Referring to the mineral's Hugoniot as described in section 2, these transformations will be accomplished generally by shock states that lie along the two-phase and high-pressure-phase regimes of the Hugoniot (Fig. 12 and 29). Whereas high-pressure-phase transitions may be predicted by the Hugoniot data, the formation of short-range-order phases may be only inferred from release adiabat measurements which reveal the P, V-states through which the compressed phase passes on pressure release and which finally yield the post-shock

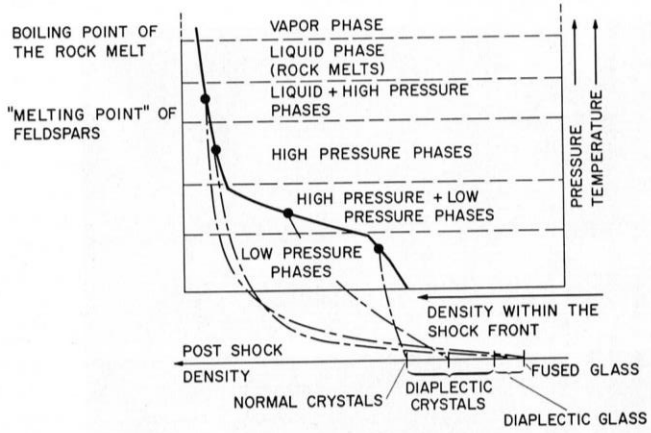


Fig. 29. Generalized Hugoniot curve of the framework silicates in the pressure-density plane showing the main regimes of the Hugoniot: low-pressure-phase regime, two-phase regime and high-pressure-phase regime. Lower scale (post-shock density) replaces the abscissa of the Hugoniot plot in order to show the structural state (post-shock density) of the recovered mineral upon pressure release. Dashed lines may help to relate the post-shock density with the corresponding shock states in the Hugoniot plot; they do not represent the location of release adiabats. Boiling and melting point refer to a qualitative post-shock temperature scale. The temperature assigned to pressure-density states which yield "liquid state" glass upon pressure release may be lower than the mineral's melting point, but higher than the softening point of the corresponding glassy phase (CHAO et al., 1970; STÖFFLER, 1971a).

Table 5. Data on shock-induced high-pressure-phases in rock-forming minerals for which dense metastable polymorphs are known from natural observations or from shock recovery experiments. hpp = high pressure phase;* density for $MgSiO_3$-composition. References in parentheses indicate data on tektites which are presumably of terrestrial origin; a = Du Pont de Nemours Co., Netherland Patent Release Nr. 6506395.

Low pressure phase (density in g/cm³)	Hugoniot high pressure phase	Metastable high pressure phase (density in g/cm³)	Pressure range for metastable hpp [kbar]	Laboratory shock synthesis (reference)	TNT and nuclear explosion craters (reference)	Occurences of metastable hpp Terrestrial impact craters (reference)	Meteorites (reference)
Quartz (2.65)	Stishovite	a) Coesite (2.93)	~ 300—500	78	65, 155, 214	36, 38, 54, 55, 59 130, 137, 145, 214, 221, 241	(229, 232)
		b) Stishovite (4.23)	~ 150—400	71	—	53, 56, 130, 214	—
Plagioclase (2.63—2.76)	„Hollandite-like" structure	Jadeite (3.24)	(> 150)	—	—	126	—
Pyroxene (3.20—3.52)	Spinel + stishovite, or „ilmenite", or „perovskite", or „garnet"-(majorite-) structure	Majorite (3.67)*	(> 135)	—	—	—	205
Olivine (3.22—4.34)	"Spinel"-or "$SrPbO_4$"-structure or periclase + stishovite	Ringwoodite (3.90)	(> 450)	—	—	—	26
Rutile (4.23)	"Fluorite"- or "$PbCl_2$"-structure	Orthorhombic TiO_2 (4.37)	> 150 (—1000)	141, 149	—	—	—
Graphite (2.25)	1) Diamond	a) Diamond (3.50)	> 130	70	—	—	115, 143, 164, 165, 231
	2) Metallic carbon	b) Hexag. diamond (3.51)	700—1400	a	—	—	115
		c) Hexag. carbon (3.43)	(> 600 ?)	—		91,92	

density (Ahrens & Rosenberg, 1968; Ahrens et al., 1969a; Petersen et al., 1970).

5.4.1. High-pressure-phases

Most rock-forming minerals undergo phase transitions to dense high-pressure-phases by dynamic compression as indicated by their measured Hugoniot data (Wackerle, 1962; McQueen et al., 1963, 1967a; Ahrens & Gregson, 1964; Wang, 1967; Anderson & Kanamori, 1968; Ahrens et al., 1969b; Trunin et al., 1971a). Little is known about the mineralogical nature of the high-pressure-phases for some minerals. Since most phase transitions of silicates are sluggish and have high activation energies, a shock transition very often yields no crystallized dense phase upon pressure release mainly because of the short duration of the pressure pulse and/or high post-shock temperatures. In the case when a high-pressure-phase actually crystallizes during the shock process and is preserved metastably, only a small proportion is generally recovered after pressure release. This dense, metastable phase or phase assemblage might be formed either by a polymorphic transition or by a breakdown into a high-pressure-phase assemblage, respectively.

High pressure phases have been recovered from naturally shocked minerals in rocks of impact structures and in meteorites (Table 5). Stishovite (de Carli & Milton, 1965), coesite (Deribas et al., 1966), orthorhombic titanium dioxide (McQueen et al., 1967b), diamond (de Carli & Jamieson, 1961) and hexagonal diamond (DuPont De Nemours patent quoted by Hannemann et al., 1967) have been synthesized by laboratory shock recovery experiments. Coesite has been found in rocks of TNT and nuclear explosions (Milton et la., 1961; Dachille et al., 1968; Stöffler, 1971b). The data on naturally and experimentally produced high-pressure-phases (see also Table 5) suggest that polymorphic transitions take place preferentially whereas the breakdown to phase assemblages that require extended diffusion processes is rarely observed. Moreover, two types of mechanisms have to be distinguished in the formation of high-pressure-phases:

(1) phases which have formed during shock compression and are characteristic of the high-pressure-phase regime of the Hugoniot curve, e.g., stishovite, diamond.

(2) phases which have presumably formed by reversion of phases of type (1) during or after pressure release, e.g., coesite, rhombic TiO_2. This process has been suggested by several authors (de Carli & Milton, 1965; McQueen et al., 1967b; Stöffler, 1971b).

The formation conditions are not well known for most of the shock-induced high-pressure-phases. The P, T-field in which these phases can be produced and metastably preserved is determined firstly by the peak pressure for the onset of the two-phase regime of the Hugoniot curve determining the lower limit of P, T-conditions and secondly by the thermal stability of the metastable phase at zero pressure which determines the upper limit of post-shock temperature — and hence of the related peak

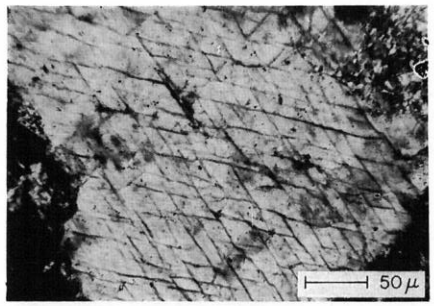


Fig. 15

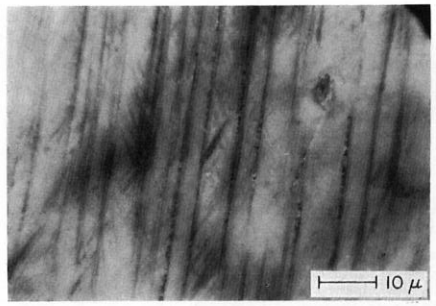


Fig. 17

Fig. 15. Planar fractures parallel to $\{10\bar{1}1\}$ in quartz of a granite from the "Bunte Breccie" of Ronheim, Ries, Germany (S 614). Crossed polarizers.

Fig. 17. Non-decorated planar elements in quartz parallel to $\{10\bar{1}3\}$ and $\{10\bar{1}2\}$ from a gneiss fragment in the crystalline breccia of Appetshofen, Ries, Germany (S 289). Some microstructures (N—S-trending) appear as "filled lamellae" containing minute crystals, probably stishovite. Crossed nicols.

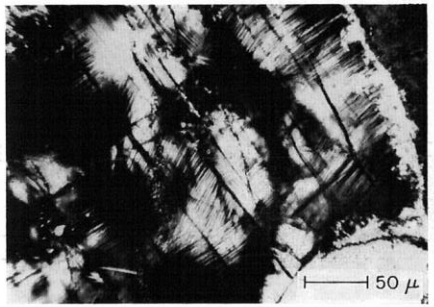


Fig. 18a

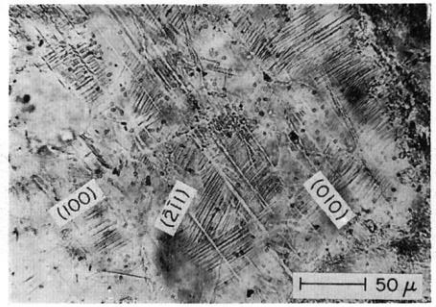


Fig. 18b

Fig. 18. Planar elements (isotropic lamellae) parallel to (100) and $(\bar{2}\bar{2}1)$, and isotropic twin lamellae parallel (010) in Andesine (An_{31}) of a gneiss from the suevite of Zipplingen Ries, Germany (S 349). Black areas = irregular isotropic regions of the crystal a) crossed nicols b) plane polarized light.

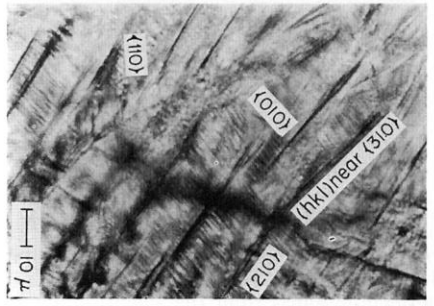


Fig. 19

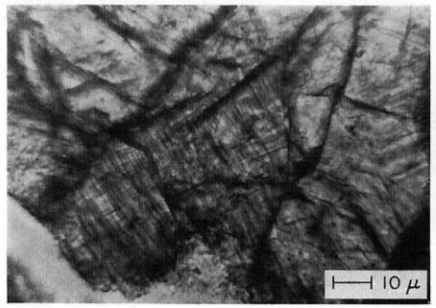


Fig. 20

Fig. 19. Planar elements in sillimanite of a garnet-sillimanite schist from the suevite of Otting, Ries, Germany (S 1051). Crossed nicols.

Fig. 20. Sets of planar elements oriented parallel to {h0l} in pyroxene from an amphibolite included in the crystalline breccia of Maihingen-Langenmühle, Ries, Germany (S 20). Plane polarized light.

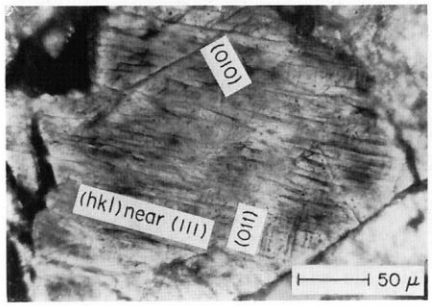


Fig. 21

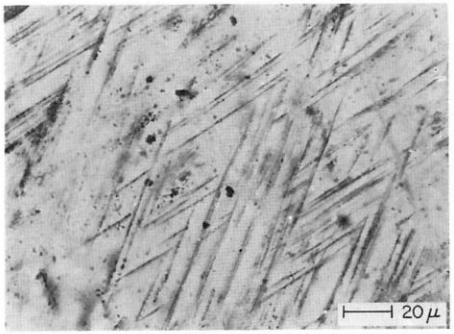


Fig. 22

Fig. 21. Sets of planar fractures and decorated planar elements in amphibole (hornblende) from an amphibolite included in the suevite of Otting, Ries, Germany (S 140). Crossed polarizers.

Fig. 22. Decorated planar elements parallel to $\{10\bar{1}3\}$ of a gneiss from suevite of Zipplingen, Ries, Germany (S 379). Plane polarized light.

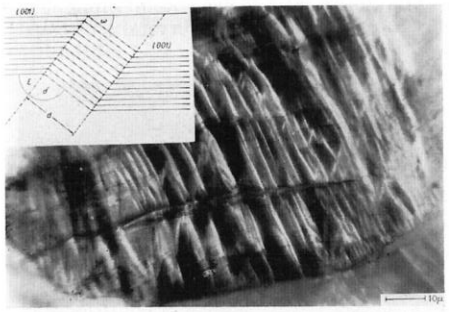


Fig. 23a

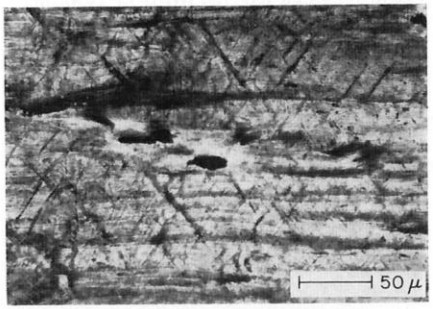


Fig. 23b

Fig. 23. a) Kink bands in biotite of a garnet-sillimanite-schist from the suevite of Otting, Ries, Germany (S 1051).

b) Planar fractures in biotite of a granite from the suevite of Aumühle, Ries, Germany, oriented parallel to $\{111\}$ and $\{1\bar{1}1\}$ (S 744).

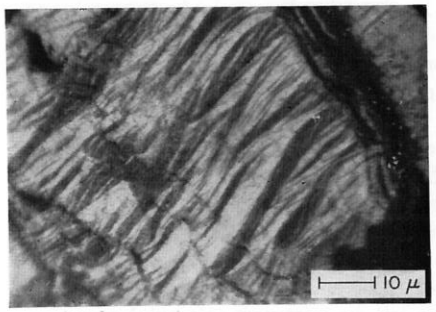


Fig. 24

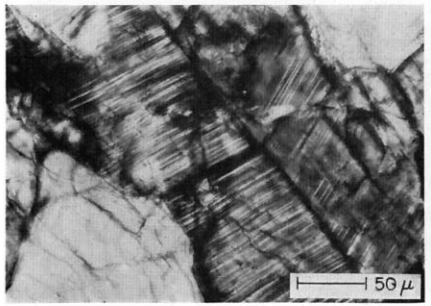


Fig. 25

Fig. 24. Kink bands in shocked graphite of a garnet-sillimanite-schist from the suevite of Otting, Ries, Germany (S 1050). Thin section approximately perpendicular to the plane (001). Plane polarized reflected light (oil immersion).

Fig. 25. Mechanical twins in lunar clinopyroxene from a basaltic rock fragment. Preshock twin plane parallel to (100), shock-induced twins subparallel to (001); Apollo 12, Oceanus Procellarum, No. 12057,14; crossed polarizers.



Fig. 26

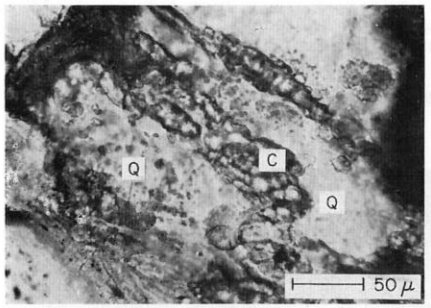


Fig. 30

Fig. 26. Mechanical twins and planar fractures in amphibole (hornblende) from an amphibolite included in the suevite of Bollstadt, Ries, Germany (B 135). Lamellae oriented near (001), N—S-fractures near {110}, NW—SE-fractures near {101}. Crossed polarizers.

Fig. 30. Coesite aggregates within diaplectic quartz glass of a granitic inclusion in the suevite of Aufhausen, Ries, Germany (S 262). C = Coesite aggregates, Q = diaplectic quartz glass.

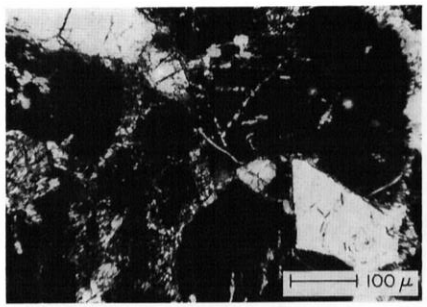


Fig. 32a

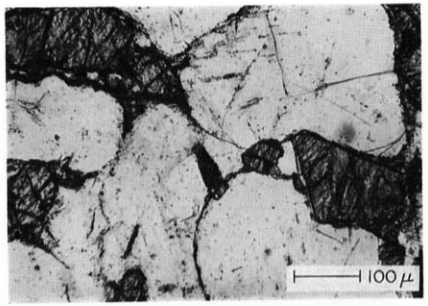


Fig. 32b

Fig. 32. Diaplectic andesine glass preserving all morphological features of the previous crystalline state. Gray grains (in b)) are hornblende (amphibolite fragment from the suevite of Alerheim, S 15). a) crossed nicols b) parallel nicols.

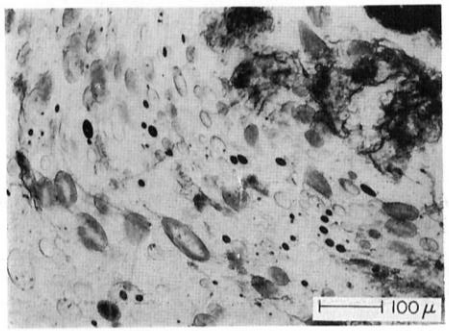


Fig. 33

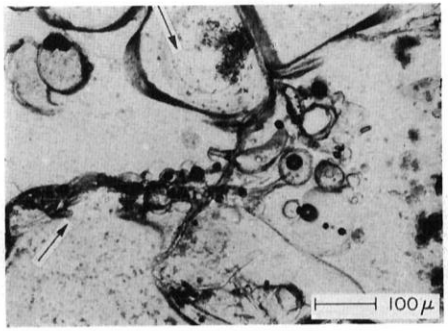


Fig. 34

Fig. 33. Shock fused alkali feldspar ($Or_{57}Ab_{40}An_3$) quenched to a glass with vesicles and flow structures from a granitic rock fragment included in the suevite of Otting (S 169). Grain of partially recrystallized diaplectic quartz in the upper right corner. Plane polarized light.

Fig. 34. Inclusion of fused vesicular quartz (lechatelierite) within the suevite of Otting, Ries, Germany. Arrows indicate areas of large vesicles.

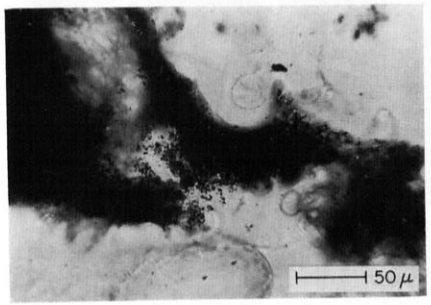


Fig. 35

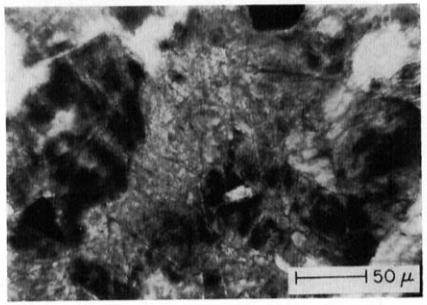


Fig. 36

Fig. 35. Very fine-grained opaque aggregates (magnetite) within oxidized biotite surrounded by vesicular alkali feldspar glass of a granitic inclusion in the suevite of Bollstadt (S 115). Ries, Germany.

Fig. 36. Opaque oxidation products (black areas) in amphibole (hornblende) from an amphibolite included in the suevite of Otting, Ries, Germany (S 8). Plane polarized light.

pressure — which can be sustained by the high-pressure-phase without being thermally decomposed after pressure release. Consequently, the mode of pressure and temperature decay, which may be very different in various shock processes, is essential in governing the pressure limit at which a dense phase will be preserved metastably.

In natural shock processes secondary thermal annealing at temperatures higher than the actual post-shock temperature may obliterate the primary formation conditions. This occurs, for example, if moderately shocked rocks are buried beneath hot layers of impact breccias (STÖFFLER, 1971b).

The actual formation rate of high-pressure-phases is also very much influenced by the duration of the peak pressure as well as by the whole shock transition time. An increase of this time obviously favors the crystallization of dense metastable phases. For example coesite does not form in small-scale laboratory shock experiments on non-porous quartzose materials but is generally present in shocked rocks of large-scale artificial explosions as well as terrestrial impact structures (see STÖFFLER, 1971b).

Available data on shock-induced high-pressure-phases of some rock-forming minerals are briefly described below (see also Table 5).

1. Quartz. Quartz transforms partially to coesite and stishovite upon shock compression (Fig. 17 and 30). Both polymorphs were found first in nature by CHAO et al. (1960, 1962) in shocked sandstones of the Arizona meteorite crater. Coesite is now known from other terrestrial impact sites, from large TNT and nuclear explosion craters (STÖFFLER, 1971b and references therein) and from tektites (WALTER, 1965; WEISKIRCHNER, 1968). Stishovite has been found only at the Arizona and Ries craters. Both polymorphs have been produced by laboratory recovery shock experiments (DE CARLI & MILTON, 1965; DERIBAS et al., 1966). Data on the quantitative abundances and the formation conditions in naturally shocked porous and non-porous rocks are available from detailed studies of the Coconino-sandstone (KIEFFER, 1971) and of crystalline rocks of the Ries crater (STÖFFLER & ARNDT, 1969; STÖFFLER, 1971b), respectively. The reader is referred to these papers for details. The formation conditions of coesite and stishovite in non-porous rocks are given in Fig. 31 and 16. They are considerably different from those in porous materials as has been shown by KIEFFER (1971).

2. Plagioclase. It has been deduced from Hugoniot data (AHRENS et al., 1969a) that feldspars might transform to a dense, hollandite-like structure (RINGWOOD et al., 1968) by dynamic compression, but a phase of this type has not yet been detected in shocked feldspars. However, JAMES (1969b) found jadeite in shocked oligoclase of the Ries crater, which probably represents a metastable quench product.

3. Olivine. No high pressure polymorph of olivine has so far recovered from shock experiments or from terrestrial or lunar shocked rocks although the Hugoniot data clearly indicate a phase transition above about 450 kbar (MCQUEEN et al., 1967a). Possible candidates for this high pressure phase are the spinel or the $SrPbO_4$-structure or an assemblage of oxides (periclase + stishovite). The spinel-phase (ringwoodite) has been discovered recently

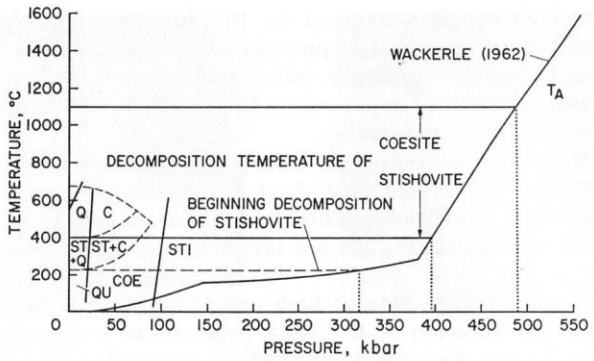


Fig. 31. Pressure-temperature conditions for the metastable preservation of coesite and stishovite during shock compression of quartz within non-porous rocks. The solid curve represents the post-shock temperature T_A in quartz calculated by WACKERLE (1962). Inclined solid lines indicate the phase boundaries of the phase equilibrium diagram of silica (*Qu* quartz; *Coe*, coesite; *Sti*, stishovite). Curved brocken lines after GIGL & DACHILLE (1968) show the various crystalline transformation products of stishovite as a function of static pressure and temperature at a reaction time of 3 days ('wet') system; Q indicates complete conversion to quartz, C indicates complete conversion to coesite, St + C indicates partial conversion to coesite and amorphous silica, St + Q indicates partial conversion to quartz and amorphous silica). Vertical dotted lines relate decomposition temperatures to peak pressures (see text).

in the Tenham meteorite (BINNS et al., 1969). Its formation is considered to be due to shock compression.

4. Pyroxene. According to the Hugoniot data of AHRENS & GAFFNEY (1971) pyroxene undergoes a phase transition above about 135 kbar. The high-pressure-phase will probably be represented by a garnet structure "majorite" (see RINGWOOD & MAJOR, 1966; Table 5). However, this phase has not yet been recovered from any shocked pyroxene. It was found in the Coorara meteorite by SMITH & MASON (1970) who did not specify the process of formation either by dynamic or static compression.

5. Rutile. It can be concluded from the Hugoniot data that rutile undergoes a phase transition at dynamic pressure above 200 kbar (LINDE & DE CARLI, 1969; MCQUEEN et al., 1967b). The same authors were able to recover an orthorhombic high-pressure polymorph with α-PbO_2-structure (see also BENDELIANY et al., 1966) from laboratory shock experiments performed at pressures in the 150—1000 kbar range. This phase might be a metastable quench product that reverted during pressure release from a still denser phase with fluorite or $PbCl_2$-structure that actually represented the polymorph formed under shock compression.

6. Graphite. Graphite transforms to a high-pressure-phase upon dynamic compression as indicated by the Hugoniot data (ALDER & CHRISTIAN, 1961; DORAN, 1963). Shock loading of graphite has resulted in the formation of two different polymorphs, diamond (DE CARLI & JAMIESON, 1961) and hexagonal "diamond" (Du Pont de Nemours Co., Netherlands Patent Release No. 6506395). Diamonds in meteorites are considered to be

of shock origin (NININGER, 1956; LIPSCHUTZ & ANDERS, 1961). Also hexagonal diamond was discovered in shocked iron meteorites (HANNEMANN et al., 1967). Another metastable polymorph of graphite, with hexagonal structure was found by EL GORESY & DONNAY (1968) in shocked rocks of the Ries crater.

5.4.2. Short-range-order phases (diaplectic glasses)

The second type of shock-induced solid state transformations, namely the formation of amorphous short-range-order phases, seems to be unique to framework silicates. First observed in experimentally shocked quartz by DE CARLI & JAMIESON (1959), iotropic and X-ray amorphous minerals of quartz, plagioclase and alkali-feldspar composition have been found to be very common shock products of terrestrial (CHAO, 1967a, b, 1968; BUNCH et al., 1967, 1968; BUNCH, 1968; BUNCH & CASSIDY, 1972; DWORAK, 1969; ENGELHARDT et al., 1967; ENGELHARDT & STÖFFLER, 1968; STÖFFLER, 1965, 1966, 1967, 1971a), lunar (e.g. CHAO et al., 1970, ENGELHARDT et al., 1970, 1971; QUAIDE & BUNCH, 1970) and meteoritic (BINNS, 1967; DUKE, 1968) impact environments. Such glasses have been produced experimentally in shock recovery experiments (DE CARLI & JAMIESON, 1959; MILTON & DE CARLI, 1963; MÜLLER & DÉFOURNEAUX, 1968; SHORT, 1968a; STÖFFLER & HORNEMANN, 1972) and nuclear explosions (SHORT, 1966, 1968b, 1969, 1970a; JAMES, 1969a; BORG, 1972). Because of their particular mode of formation as solid state glasses they have been called "diaplectic glasses" (ENGELHARDT et al., 1967; ENGELHARDT & STÖFFLER, 1968). CHAO (1967b) introduced "thetomorphic glass" as a virtually synonymous term. The term "maskelynite" has been used for solid state plagioclase glasses (TSCHERMAK, 1872; BUNCH et al., 1967).

Diaplectic glasses are generally considered as reversion products of the corresponding high-pressure-phases which were formed during compression at shock states along the high-pressure-phase regime of the Hugoniot curve (AHRENS & ROSENBERG, 1968; AHRENS et al., 1969a; STÖFFLER, 1969, 1971a; Fig. 29). Their mode of formation during pressure release has been unmistakenably demonstrated by release adiabat measurements of AHRENS and coworkers (loc. cit.) which indicate that during pressure decay the dense high-pressure-phase suddenly transforms to a phase with very low post-shock density below about 50 kbar. Diaplectic glasses are produced within a certain pressure range, the upper limit of which is determined by the magnitude of the post-shock temperature (Fig. 16 and 29). This temperature must be lower than the melting point or at least the glass transition temperature of the mineral phase in question (Fig. 29). At higher temperatures, implicating higher pressures, shock fusion takes place and glasses quenched from a liquid will be formed. Diaplectic glasses are distinctly different in physical properties and structural state from those quenched from a liquid phase (ENGELHARDT et al., 1967; CHAO, 1967b, 1968; BUNCH et al., 1967, 1968; DUKE, 1968; ENGELHARDT & STÖFFLER, 1968; STÖFFLER & HORNEMANN, 1972, Part II).

Diaplectic glasses of quartz and feldspar composition appear in shocked crystalline rocks as isotropic phases which have preserved all morphological

features of the previous crystalline state, e.g., grain and twin boundaries (Fig. 32). In dense crystalline rocks diaplectic quartz glass generally coexists with coesite and minor amounts of stishovite unless decomposed by thermal metamorphism (STÖFFLER, 1971b). However, in porous rocks (sandstones) small amounts of diaplectic glass coexist with fused and crystalline quartz as well as coesite and stishovite due to very strong pressure and temperature heterogeneties characteristic of porous rocks (KIEFFER, 1971). It can be concluded from recovery shock experiments (HÖRZ, 1968; MÜLLER & DÉFOURNEAUX, 1968; MÜLLER & HORNEMANN, 1967; KLEEMANN, 1971; STÖFFLER & HORNEMANN, 1972) that the formation of diaplectic quartz and feldspar glasses begins at peak pressures of about 300 kbar and is completed above 350—380 kbar though quartz seems to be slightly more resistant than feldspar. The upper pressure limit for the formation of diaplectic quartz and feldspar glasses may be seen from Fig. 16.

5.5. Thermal Decomposition, Melting and Vaporization

Shock-induced thermal decomposition, melting or even vaporization may result on pressure release from all shock states — generally located along the high-pressure-phase regime of the Hugoniot curve — generating post-shock temperatures in excess of the decomposition, melting or vaporization temperature of the mineral in question (see Fig. 29). If short-range-order phases are produced during the shock compression, then lower post-shock temperatures exceeding the softening point of the corresponding glass may be sufficient to produce the same physical effects in shocked minerals which are characteristic of the glass formation by quenching. There is certainly a gradual transition between the formation of diaplectic glasses and glasses formed by this type of thermal process.

5.5.1. Shock melting and vaporization

Shock melting of quartz, plagioclase, alkali feldspar and the formation of monomineralic glasses has been widely observed in shocked terrestrial rocks (SPENCER, 1933; SHOEMAKER & CHAO, 1961; STÖFFLER, 1966, 1967; CHAO, 1967a, b, 1968; ENGELHARDT, 1967; ENGELHARDT & STÖFFLER, 1968; FRENCH et al., 1970; KIEFER, 1971; BUNCH & CASSIDY, 1972) but was rarely found in lunar rocks though shock fused pyroxene has been found (ENGELHARDT et al., 1970). Shock melting of the framework silicates is also common in nuclear explosion craters (SHORT, 1968b, 1969, 1970a; JAMES, 1969a). The conditions of the formation of monomineralic glasses in rocks are highly dependent on the mutual differences between the melting points of the coexisting phases. If these differences are high enough, selective melting of a particular phase may be achieved as frequently observed for feldspar in granitic rocks where fused feldspars commonly coexist with solid diaplectic quartz glass and oxidized biotite. The formation of mixed silicate glasses is favored and finally predominant if the differences between the melting points of the coexisting phases, e.g. pyroxene and anorthite in lunar basalts, decrease and the peak pressure and hence the post-shock temperature increase.

Shock fused minerals are characterized morphologically by flow structures and vesiculation (Fig. 33, 34) although this might not be a valid criterion of shock fusion alone. However, other physical properties are indicative of the specific thermal history of such glasses and yield reliable criteria for the distinction between those and solid state glasses (see Part II, STÖFFLER, 1972).

Only few data on peak pressures required for shock melting of single crystals are available (Table 6). Shock fusion of quartz was achieved by peak pressures between 500 and 600 kbar (DE CARLI & JAMIESON, 1959; WACKERLE, 1962; MÜLLER & HORNEMANN, personal communication). In a series of shock recovery experiments we produced shock melting of orthoclase, oligoclase and labradorite at pressures between 475 and 575 kbar by multiple shock reverberations (STÖFFLER & HORNEMANN, 1972). Local vesiculation was observed in oligoclase and labradorite glasses formed at pressures of 425 kbar. MILTON & DE CARLI (1963) fused bytownite at 600 kbar. The minimum pressure for the shock fusion of feldspar by a single shock is certainly lower (Fig. 16) because a single shock produces more entropy.

Little is known about the residual effects of shock vaporization. If a vapor phase is produced, this phase may yield condensation products on cooling such as liquid droplets solidifying to glassy bodies of revolution, mostly spherules. Glassy and metallic spherules of polymineralic composition are very common in lunar detrital impact formations (LEVINSON, 1970, 1971). Part of those could have been formed by condensation of a vapor phase. There are no criteria as yet, however, which make it possible to distinguish between droplets of glass or metal formed by condensation and those formed directly from a spray of liquid.

Very recently AHRENS & O'KEEFE (1972a) made a detailed thermodynamic study of shock melting and vaporization of rock-forming minerals. They calculated the entropy along the Hugoniot curve and compared the obtained results with the entropies required to melt or vaporize a mineral at ambient pressures. These calculations were adapted to the process of hypervelocity impact of iron and aluminum projectiles simulating natural meteoroid impact into non-porous and porous materials. AHRENS and O'KEEFE found that iron projectiles will produce melting of silicates at velocities of ~ 4 to 6 km/sec increasing in the order quartz, plagioclase, olivine, and pyroxene (see also VEDDER, 1971). Incipient vaporization by impacting aluminum projectiles requires impact velocities of ~ 7 to 11 km/sec. This velocity decreases markedly with increasing porosity of the target. Complete melting of particulate material of basaltic composition with a bulk density of 1.3 to 2.3. g/cm³ is expected at impact velocities of 3 and 4 km/sec for iron and aluminum projectiles, respectively. This fact helps to explain the great amount of glass found in the detrital regolith at the lunar surface.

5.5.2. Thermal decomposition

A number of common rock-forming minerals such as amphibole and biotite do not undergo congruent melting when exposed to high post-shock temperatures but rather decompose to an assemblage of phases (Fig. 35

Table 6. Shock-induced melting and thermal decomposition of rock-forming minerals; (1) = references see Table 1; references in parentheses indicate data on tektites which are presumably of terrestrial origin.

Mineral	Post-shock phase(s)	Formation pressure [kbar]	Laboratory shock production	TNT and nuclear explosion craters (1)	Occurences in natural rocks: Terrestrial impact craters	Occurences in natural rocks: Lunar rocks	Meteorites
Quartz	Glass	> ~ 500	69, 160, 217, 227	Sedan, Scooter, Hardhat	38, 50, 51, 52, 54, 92, 130 152, 196, 206, 208, 217, 243		(22, 48)
Plagioclase	Glass	> ~ 425—450	154, 160, 199, 217	Danny Boy, Hardhat	38, 50, 52, 208, 209, 217	175	
Alkali feldspar	Glass	> ~ 450	160, 217	Sedan, Hardhat	38, 50, 51, 52, 208, 210, 217, 243		
Olivine	Iron oxides + SiO_2 or recrystallization	> ~ 600	45, 191	Danny Boy		45	
Pyroxene	Glass or iron oxides + SiO_2 or recrystallization	> ~ 550—600		Danny Boy	212	100	
Amphibole	Magnetite + amorphous phases	> ~ 500—550			1, 50, 51, 52, 208, 210		
Biotite	Magnetite + herzynite + pyroxene + feldspar + amorphous silicate	> ~ 450—500	190	Hardhat, Sedan	38, 50, 51, 52, 190, 208, 210		
Zircon	Baddeleyite + SiO_2				88, 89		
Rutile	Quenched melt				87, 89		
Magnetite	Quenched melt or hematite	> ~ 600—700			87, 89		
Ilmenite	Rutile + pseudobrookite				87, 89		

and 36). This type of shock-induced thermal decomposition is frequently observed in minerals of highly shocked rocks such as in biotite, amphibole, olivine, magnetite, ilmenite, sphene, and zircon (CHAO, 1968; EL GORESY, 1964, 1968; STÖFFLER, 1966, 1969; KLEINMANN, 1969; SCHNEIDER, 1971). Data taken mainly from observations on terrestrial impact structures and nuclear explosion craters are summarized in Table 6.

The thermal breakdown of these minerals may be a useful indicator of the magnitude of residual temperature induced by any shock process. However, uncertainties exist until now owing to lack of Hugoniot as well as shock and post-shock temperature data of most of these minerals and of data on the possible influence of the shock pressure on the decomposition temperatures. The thermal behavior of biotite and amphibole in shocked crystalline rocks has been studied in detail (CHAO, 1968; SCHNEIDER, 1971). Both minerals are gradually oxidized at pressures exceeding an estimated lavue of 450 to 500 kbar

Biotite transforms to oxybiotite and gradually looses pleocroism and birefringence until it finally becomes opaque (Fig. 35). Hornblende, as observed in Ries rocks, becomes dull-brownish in appearance and gradually opaque within patchy crystal areas (Fig. 36). Similar alteration processes of olivine and augite have been reported by SHORT (1969) and SCLAR (1969) from a nuclear explosion and from laboratory shock experiments, respectively. CARTER et al. (1968) and JAMES (1969a), however, observed mainly recrystallization of these minerals as a most typical reaction in response to shock-induced high temperatures. The properties of gradually decomposed, shocked minerals will be presented in more detail in Part II of this article (STÖFFLER, 1972).

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Erratum

1. On page 99: delete the word „metastable“ on the third line. The described phase is a stable polymorph.
2. In Fig. 14 some poles to crystallographic planes are mislabeled: (311) and (131) must read (131) and (311) respectively (compare corresponding data of Lehtinen, 1973)